

Grid-Connected Converters

Descripción



PONTIFICIA
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DE CHILE



Power and Energy Conversion Laboratory

Research in Power Electronics



Lecture Sections **Grid-Connected Converters**

1. Grid Synchronization
2. Operation Modes of Grid-Connected Converters



Compilation of base papers

- Following a thorough search on the IEEE Xplore website, **7 prestigious papers** studying Grid Synchronization and Grid Connected Converters in its various fields have been shortlisted.
- These papers are used both to present explicit information and to extract concepts and generate our own developments.



- 1) 2006 Overview of Control and Grid Synchronization for Distributed Power Generation Systems
- 2) 2007 Decoupled Double Synchronous Reference Frame PLL for Power Converters Control
- 3) 2010 Power-Synchronization Control of Grid-Connected Voltage-Source Converters
- 4) 2011 Multiresonant Frequency-Locked Loop for Grid Synchronization of Power Converters Under Distorted Grid Conditions
- 5) 2011 A review on synchronization methods for grid-connected three-phase VSC under unbalanced and distorted conditions
- 6) 2021 Grid-Forming Converters Control Approaches, Grid-Synchronization, and Future Trends—A Review
- 7) 2025 Grid-Supporting Renewable Energy Systems With Power Electronics Interfaces

Compilation of base papers



Key content covered in the selected papers:

- [1] Overview of DPGS control and grid synchronization; review of PV/FC/WT converter topologies; comparison of dq , $\alpha\beta$, and abc control frames; importance of PLL robustness under unbalanced and distorted grids.
- [2] Introduction of DDSRF-PLL for positive-sequence extraction; dual synchronous reference frames with decoupling to eliminate 2ω oscillations; fast and accurate synchronization in unbalanced/distorted grids.
- [3] Power-synchronization control (PSC) for VSCs as an alternative to PLL; SM-like synchronization behavior; improved stability in weak grids; seamless switching with current-limiting control under faults.
- [4] Introduction of MSOGI-FLL: multi-SOGI harmonic decoupling network enabling frequency-adaptive extraction of positive/negative sequences and multiple harmonics; robust synchronization under highly distorted/unbalanced grids with fast, accurate real-time implementation.
- [5] Comprehensive review of VSC synchronization methods; classification of open-loop vs closed-loop and single- vs three-phase schemes; performance limits under harmonics, unbalance, and frequency variations.
- [6] Review of grid-forming converters (GFM); unified control architecture; comparison of droop, PSC, VSM, and VOC; synchronization without PLL; discussion of stability, current limitation, and future grid-code requirements.
- [7] Review of RES grid-support functions; WT/PV/BSS ancillary services (voltage, frequency, LVRT); control strategies (droop, VSG, PR/MPC/SMC); coordination methods (decentralized, distributed, OPF-based).

Section 1: Grid Synchronization

Power Electronic Control Fundamentals



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1.1: Grid Synchronization

Fundamentals



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Synchronization overview

Grid Synchronization: purpose and relevance [1, 5]

- Provides estimates of phase (θ), frequency (ω) and magnitude (V) at PCC (*Point of Common Coupling*), enabling stable transformations to $\alpha\beta$ and dq frames and precise active/reactive power control in grid-tied power converters.
- Inadequate synchronization leads to current loop errors, incorrect P/Q injection, and potential instability, especially in weak grids.
- Why it's challenging?: Grids present harmonics, unbalance, notches, frequency drift; synchronization algorithms must be both fast and robust.
- Key techniques: Phase-Locked Loop (PLL), Frequency-Locked Loop (FLL), Second-Order Generalized Integrator (SOGI)-based QSG, Decoupled Double Synchronous Reference Frame PLL (DDSRF-PLL), and hybrid FLL-assisted PLL.

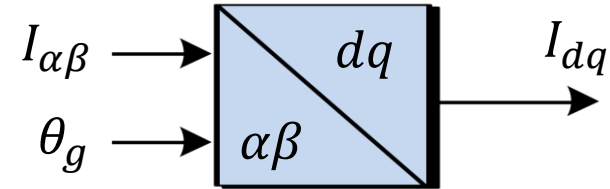


Figure 1.1. Current transformation scheme frame $\alpha\beta$ to dq : a key step for decoupled current control and active/reactive power control in grid-connected converters.

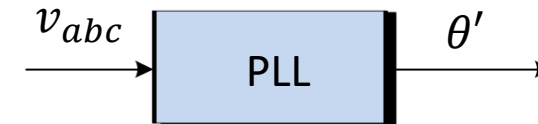


Figure 1.2. Phase-Locked Loop Simple Representation.



Figure 1.3. Frequency-Locked Loop Simple Representation.

Key Transformation: Clarke ($\alpha\beta 0$ frame)

- The Clarke transform converts the time-domain components of a three-phase system (abc frame) into two components of an orthogonal stationary frame ($\alpha\beta$ frame); it also includes a “zero” component based on the three phases (null in balanced systems). [8]
- Used by many open/closed-loop synchronizers (e.g., SOGI-based, Kalman) for clean sinusoidal templates. [5]
- The conversion expression for $abc \rightarrow \alpha\beta 0$ is given by:

$$\begin{bmatrix} \alpha \\ \beta \\ 0 \end{bmatrix} = k \cdot \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

- The change in reference frame is determined by the matrix, such that the α component is in phase with the a component.
- The factor “ k ” is an important constant for different equivalences:
 - $k = \frac{2}{3}$: Amplitude invariant transformation, $\alpha\beta$ component have the same amplitude of abc components.
 - $k = \sqrt{\frac{2}{3}}$: Power invariant, for equivalence in power expressions

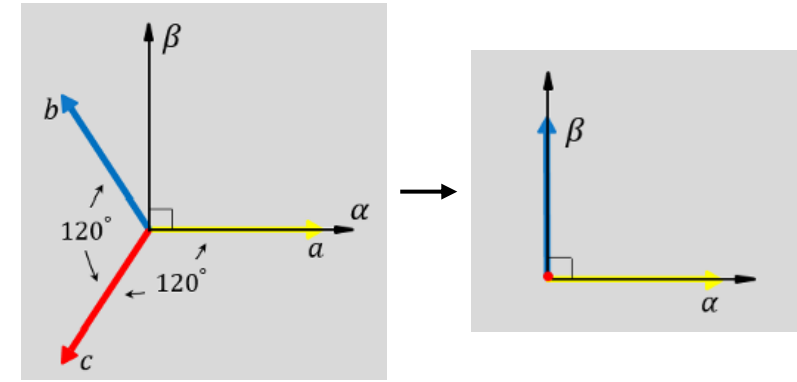


Figure 1.4. Vectorial Clarke change in Frame Representation. [8]

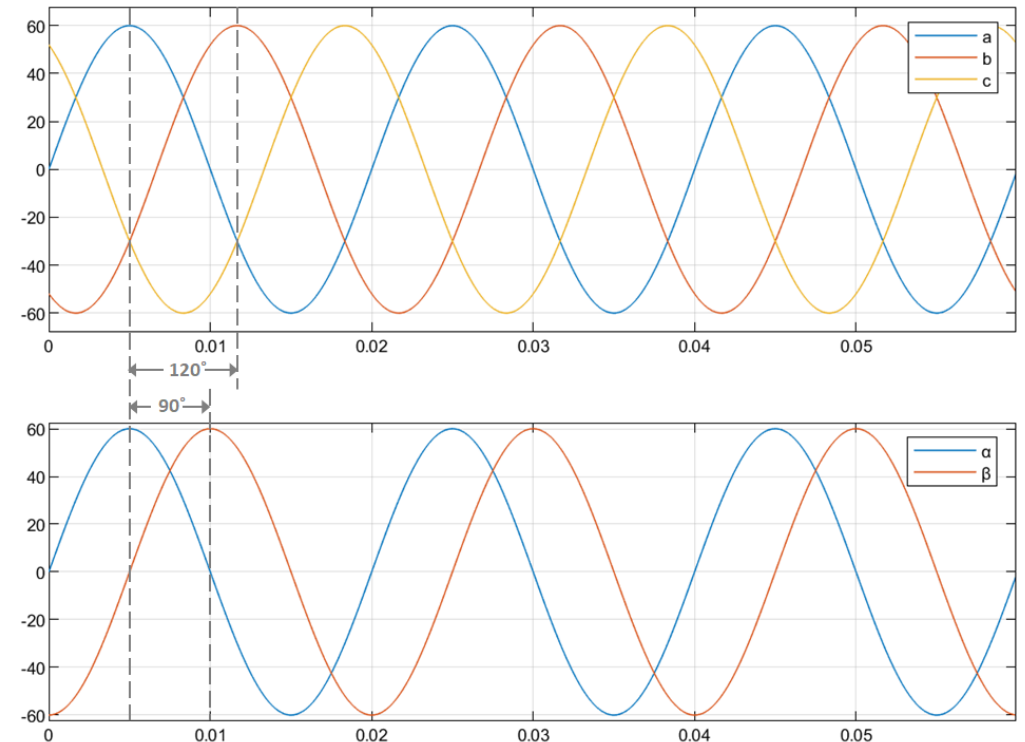


Figure 1.5. Sinusoidal signal abc and $\alpha\beta$. [8]

Key Transformation: Park (dq frame)

- Park's transformation converts the two components of the orthogonal $\alpha\beta$ frame into an orthogonal rotating reference frame (dq). $abc/\alpha\beta \rightarrow dq$ using estimated angle θ ; turns AC into DC components under steady state. [9]
- Basis of PI control of converters, the SRF-PLL and enhanced PLLs. [1, 5]
- The conversion expression for $\alpha\beta \rightarrow dq$ is given by:

$$\begin{bmatrix} d \\ q \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

- When the angle match correctly, d component is a constant with same amplitude of α component and q component is zero. [2]

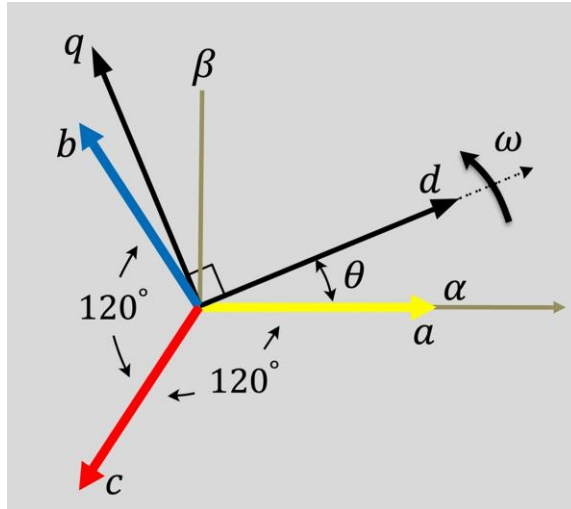


Figure 1.6. Reference frames abc , $\alpha\beta$ and dq . [9]

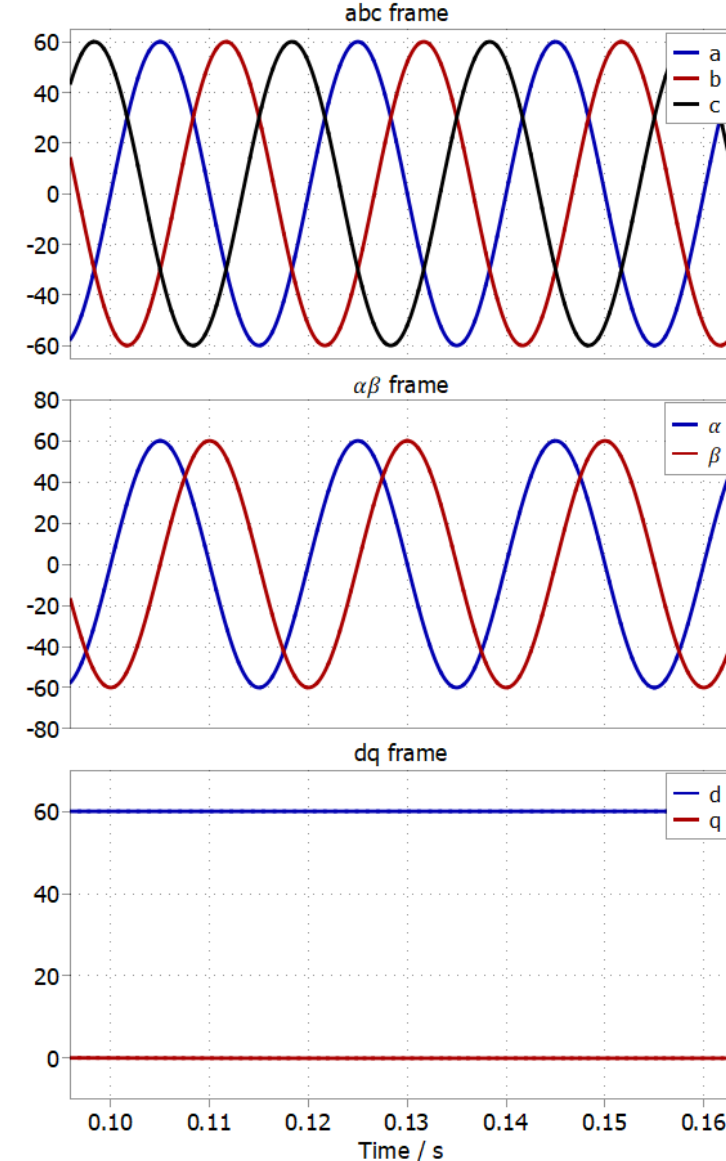


Figure 1.7. Signal representations abc , $\alpha\beta$ and dq frames.

Measurement Filters

Low Pass Filter (LPF) [1, 5]

- **Purpose:** Attenuate broadband high-frequency noise and harmonics on measured signals used by synchronizer or controller.
- **Placement:**
 - **SRF path:** filter v_d/v_q before the PI of the PLL.
 - **$\alpha\beta$ path:** pre-conditioning of measured v_α/v_β (SOGI/PR loops often embed LPF behavior).
- **Benefits:** Improves noise immunity, reduces ripple in high frequencies without adding multiple narrow notches.
- **Trade-offs:** Lower cutoff = more delay, potentially slower locking and reduced disturbance rejection bandwidth; must balance with desired dynamics.

- Transfer Function:
$$H_{LPF}(s) = \frac{1}{1 + s \cdot \frac{1}{\omega_o}}$$

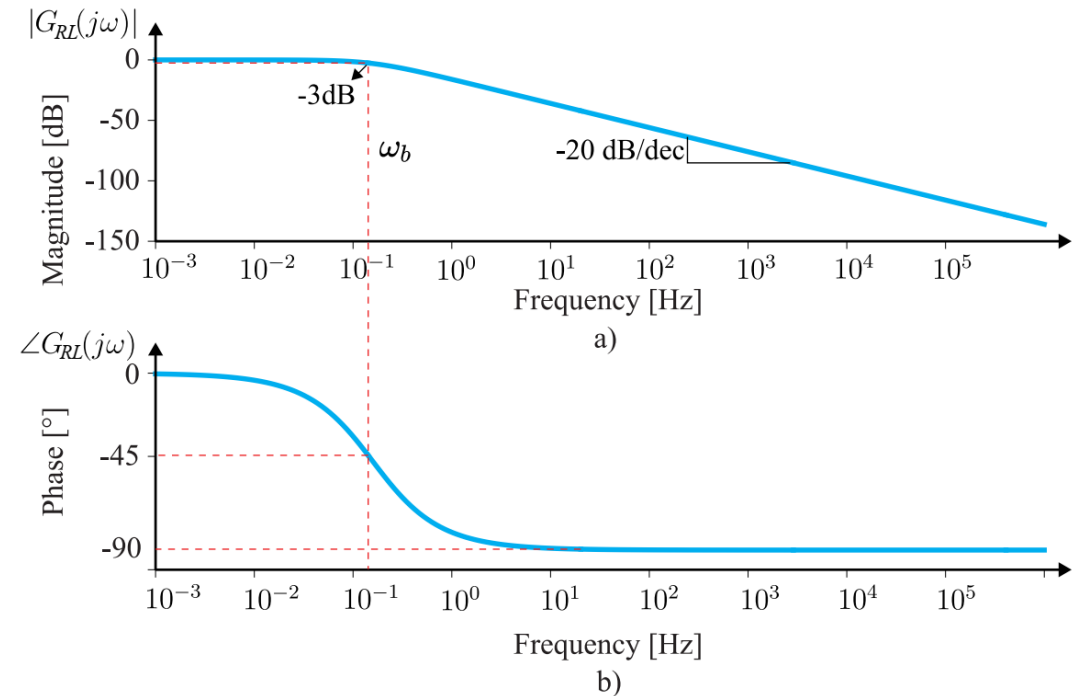


Figure 1.8. First order low Pass filter bode plot.

Measurement Filters

Notch Filter (Notch) [1, 4]

- **Purpose:** Remove specific frequency component (e.g., 2ω , 5th, 7th) that leak into PLL error signals.
- **Placement:** In rotating frame (before PI in SRF-PLL) or stationary frame ($\alpha\beta$). Usually for filter ω and 2ω components.
- **Benefits:** Strong attenuation of targeted disturbance.
- **Trade-offs:** Phase lag and slower transient response; careful tuning required vs. resonant/MAF options.

- Transfer Function:
$$H_{notch}(s) = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

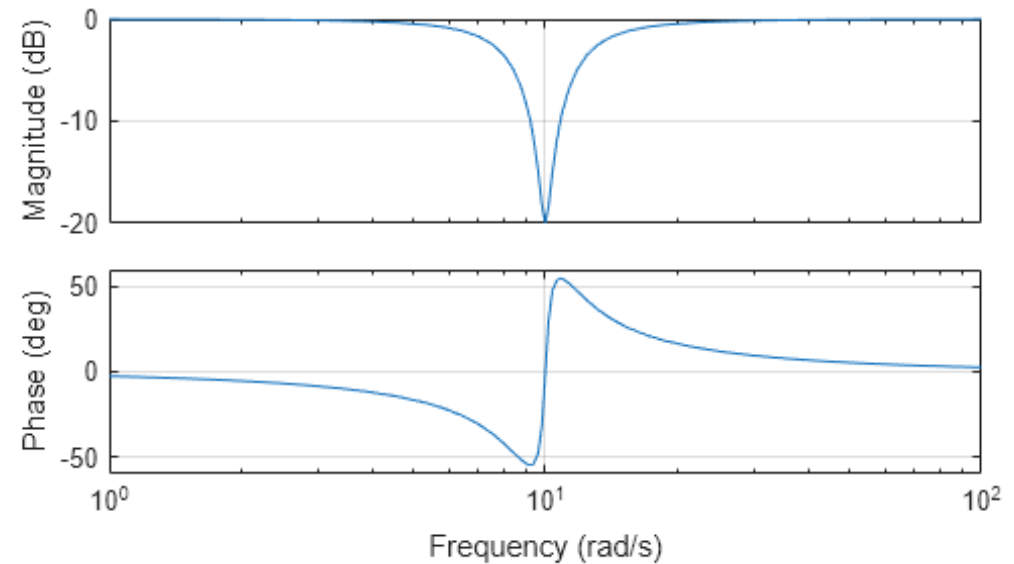


Figure 1.9. Notch filter Bode plot.

Sequence components

Definitions (three symmetrical sequences) [1,2,10]

Any set of three unbalanced phase quantities can be represented as the sum of three symmetrical sequences, according to Fortescue's symmetrical components principle:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} V_{a+} \\ V_{b+} \\ V_{c+} \end{bmatrix} + \begin{bmatrix} V_{a-} \\ V_{b-} \\ V_{c-} \end{bmatrix} + \begin{bmatrix} V_{a0} \\ V_{b0} \\ V_{c0} \end{bmatrix}$$

- **Positive-sequence (V^+):** three phasors balanced, same magnitude, $+120^\circ$, rotate at $+\omega$ (same as grid). Carries “normal” power flow and is the desired frame for control.
- **Negative-sequence (V^-):** three phasors balanced, -120° , rotate at $-\omega$; in dq produces 2ω ripple and torque/power oscillations; must be rejected for clean estimate of θ and V in synchronization loops.
- **Zero-sequence (V^0):** in-phase components in all phases; appears in systems with neutral/ground return; does not contribute to line-to-line voltage but can stress DC-link/neutral paths/transformer windings.

Why separation matters [1,2,5,6]

- Clean V^+ enables robust synchronization, PQ control, and *Fault Ride Through* (FRT) without double-frequency artifacts.
- Foundational for advanced PLLs and FLL-based synchronizers.

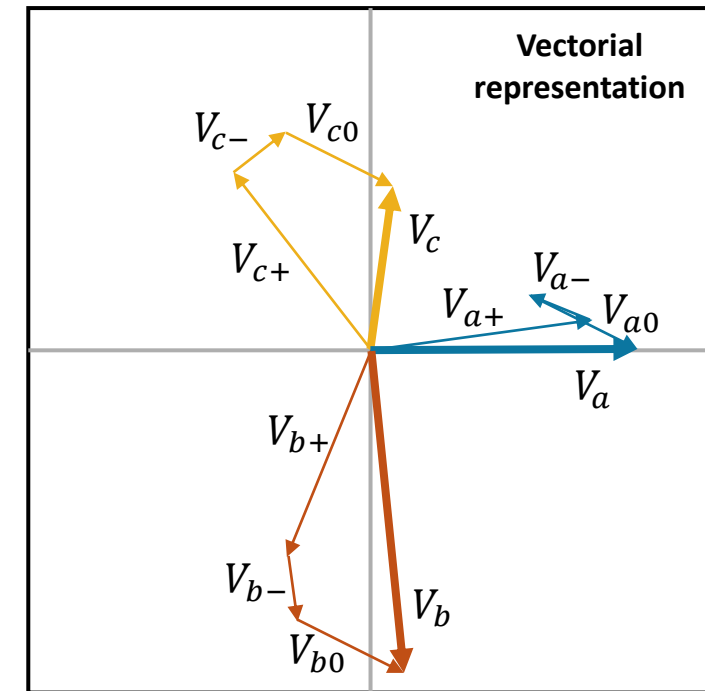
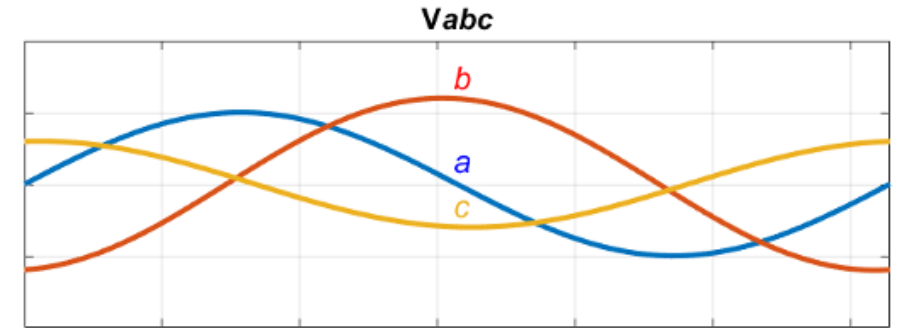


Figure 1.10. Unbalanced grid and vectorial system representation with symmetrical sequences decomposition. [10]

Sequence components

Mathematical formulation (phasor form):

- Using the complex rotation operator: $a = e^{j\frac{2\pi}{3}}$
- The symmetrical components are obtained as:

$$\begin{bmatrix} V_{a+} \\ V_{a-} \\ V_{a0} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & a & a^2 \\ 1 & a^2 & a \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

- This decomposition is exact in steady state and forms the basis for sequence extraction in real-time synchronizers.
- The remaining sets of symmetrical phasors are given in terms of simple rotations of the first set:

$$\begin{aligned} V_{b+} &= a^2 V_{a+}, & V_{c+} &= a V_{a+} \\ V_{b-} &= a V_{a-}, & V_{c-} &= a^2 V_{a-} \\ V_{a0} &= V_{b0} = V_{c0} \end{aligned}$$

Inverse Transform

- The symmetrical-components transform is unique and invertible:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ a^2 & a & 1 \\ a & a^2 & 1 \end{bmatrix} \begin{bmatrix} V_{a+} \\ V_{a-} \\ V_{a0} \end{bmatrix}$$

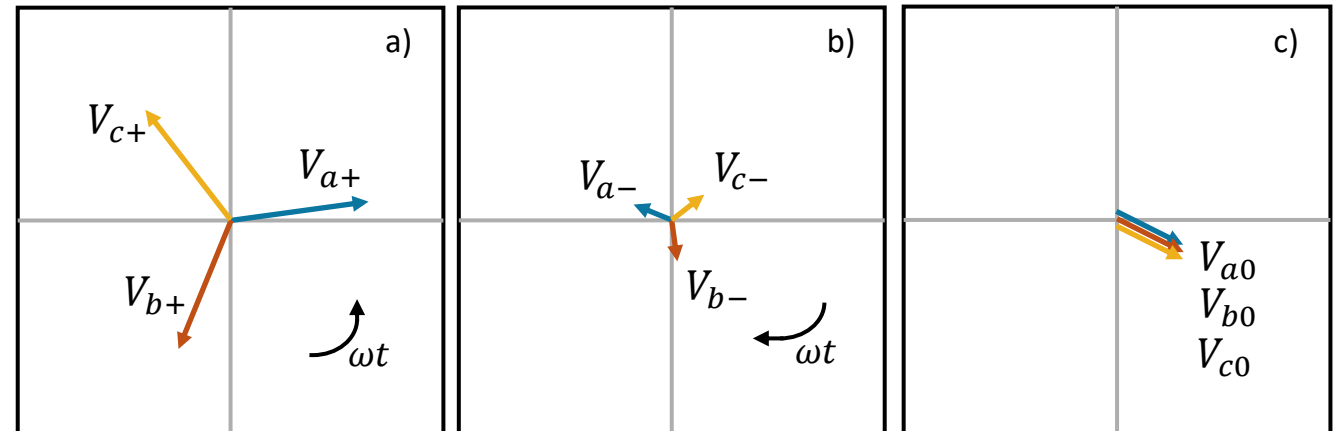
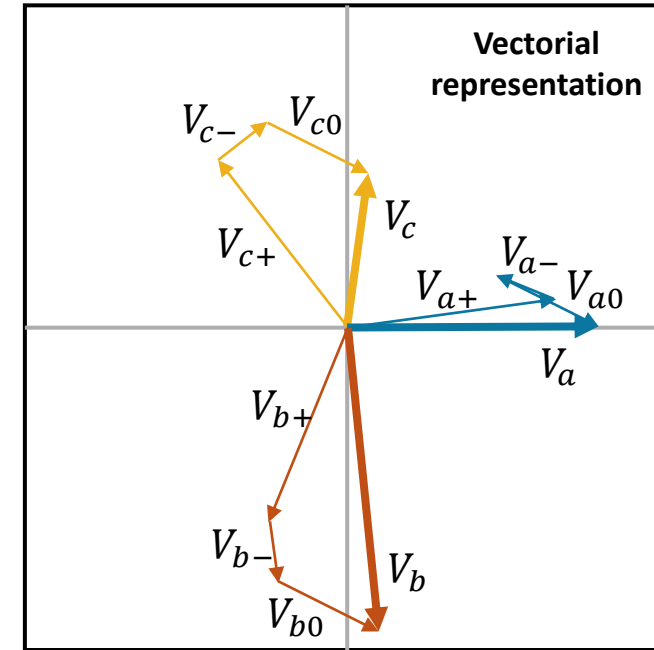


Figure 1.11. Vectorial representation of unbalanced system and symmetrical sequences decomposition. a) Positive. b) Negative. c) Zero

Synchronization methods

Overview [1,5,11]

- Grid-tied converters require accurate phase, frequency, and magnitude estimation for stable $\alpha\beta/dq$ transformations and reliable P/Q control.
- Modern grids are often distorted, unbalanced, noisy, and may experience frequency drift, so synchronization must be robust.
- Key performance needs:
 - Distortion & noise immunity
 - Frequency adaptability
 - Phase-jump tolerance
 - Unbalance robustness
 - Fast convergence
 - Simplicity & implementability

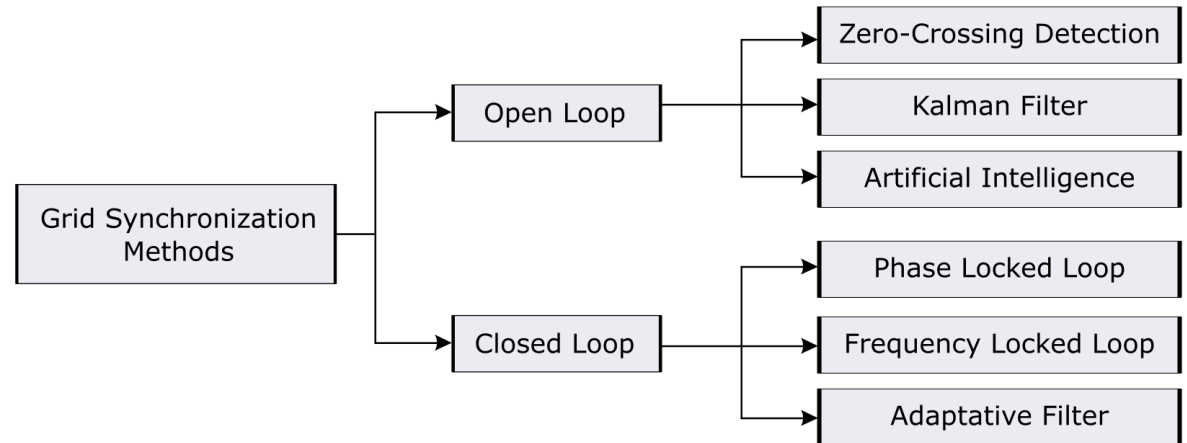


Figure 1.12. Classification of synchronization methods. [11]

Classification

Synchronization techniques are broadly grouped into **open-loop** and **closed-loop** structures, depending on whether they use feedback to refine the phase/frequency estimate.

Synchronization methods

Open-loop methods

- Estimate phase/frequency directly from voltage measurements (no feedback).
- Simple, low computational cost.
- Suitable only for ideal or mildly distorted grids.
- Examples:
 - Zero-crossing detection — very sensitive to distortion.
 - Kalman filters — accurate but computationally heavy.
 - AI-based methods — require training datasets.

Closed-loop methods

- Use feedback to adjust phase/frequency to match grid values.
- Offer higher accuracy, disturbance rejection, and stable behavior under distortion.
- Preferred for modern grid-connected converters.
- Families include:
 - PLL-based (SRF-PLL, EPLL, SOGI-PLL, DSOGI-PLL, DDSRF-PLL)
 - FLL-based (SOGI-FLL, MSOGI-FLL)
 - Adaptive filters for enhanced harmonic robustness

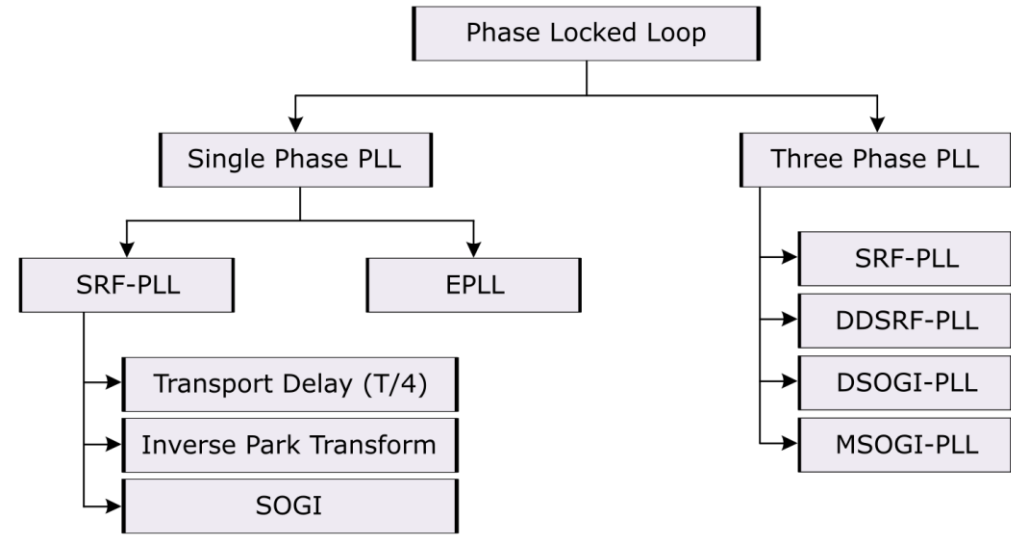


Figure 1.13. Classification of Phase Locked Loop methods.

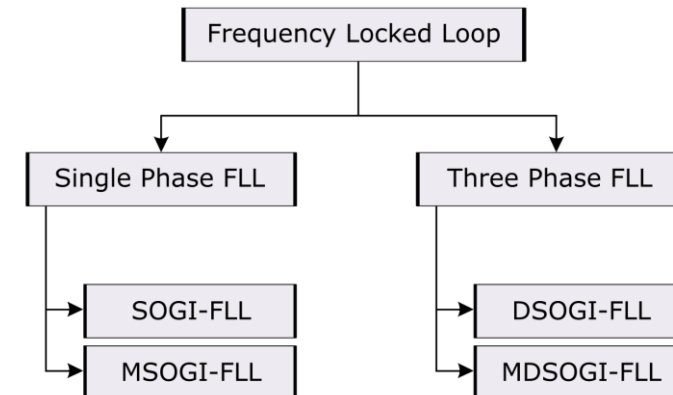


Figure 1.14. Classification of Frequency Locked Loop methods.

General structure of Phase Locked Loop (PLL)

Definition [1,5,11]

- A PLL synchronizes a converter by estimating the grid phase angle θ such that the internal reference frame tracks the grid voltage.
- A PLL locks the converter's internal angle θ to the grid phase by forcing $v_q \rightarrow 0$ in the rotating reference frame. [1]
- It is composed of three main blocks: phase detector, loop filter, and voltage-controlled oscillator (VCO/NCO).

How It Works

- **Phase detector:** compares the measured grid voltage V_g with the internally generated reference to produce the phase error ε .
- **Loop filter (usually PI):** processes ε to generate the estimated angular frequency ω' .
- **Oscillator (VCO/NCO):** processes ω' to obtain the estimated angle θ' , usually an integrator.
- The angle is fed back to the phase detector, closing the loop and ensuring continuous phase alignment.

Key Characteristics

- **Closed-loop behavior** enables adaptive tracking of phase/frequency under distortion or drift.
- **Fast and simple implementation**, widely used in grid-tied converters. [1,3]
- **Sensitive to distortion and unbalance**, requiring pre-filtering or enhanced PLL architectures. [2,5]

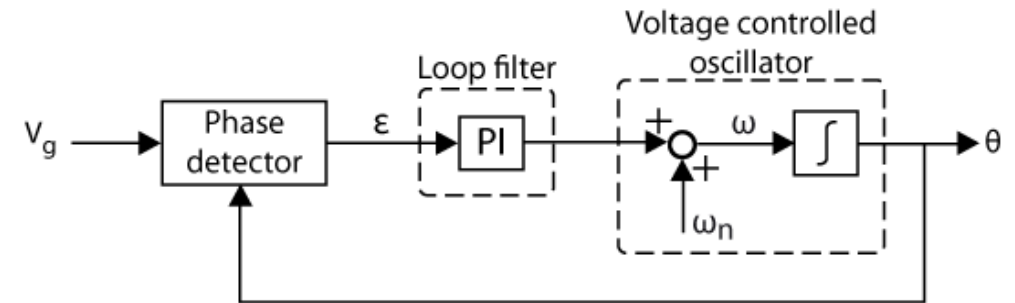


Figure 1.15. General (theoretical) structure of a Phase-Lock Loop (PLL).
[5, 11]

General structure of Phase Locked Loop (PLL)

PLL for Three Phase Systems

- Implementation is relatively straightforward as the availability of three distinct phase signals enables a direct transformation to the synchronous reference frame (dq frame), facilitating accurate phase tracking.

PLL for Single Phase Systems

- Implementing PLL-based systems in single-phase systems presents a greater challenge: how to obtain the q component.
- Various techniques have been proposed to reconstruct a pseudo-orthogonal signal (typically the β -axis) from the single-phase input (equivalent to α -axis component).
- Once $\alpha\beta$ components are obtained, Park transform can be used for calculate dq frame and apply similar PLL strategies as the ones used in three-phase systems.
- ❖ In next slides, various PLL methods for single phase and three phase systems are presented.

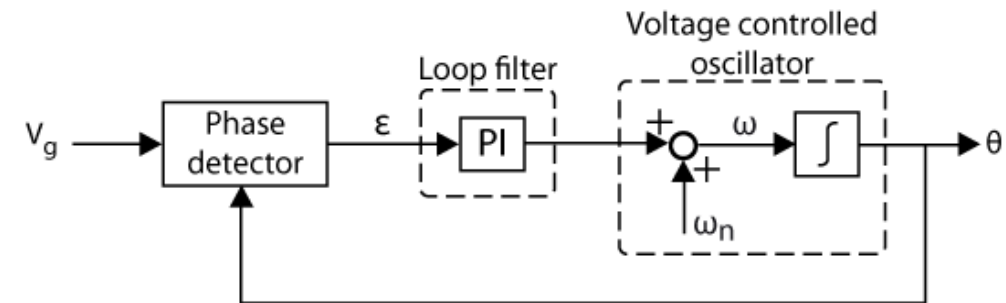


Figure 1.15. General (theoretical) structure of a Phase-Lock Loop (PLL).
[5, 11]

1.2: Grid Synchronization

Single Phase PLL



Synchronous Reference Frame PLL (SRF-PLL)

PLL methods for single phase systems

Concept [5,11]

- The SRF-PLL recreates a pseudo-quadrature signal via a Quadrature Signal Generator (QSG), enabling a Park-like transformation.
- Then, the PLL uses v_q as the phase error to align the internal reference frame with the fundamental component of the single-phase voltage.

How it works [5,11]

- **Quadrature Signal Generation:** produces $v_\beta = QSG(v_\alpha)$.
- **$\alpha\beta \rightarrow dq$ transformation:** uses estimated angle θ' to compute v_q .
- **Phase error regulation:** PI loop drives $v_q \rightarrow 0$.
- **VCO integration:** generates updated θ' .

Strengths

- Very effective under balanced sinusoidal grids.
- Simple implementation; compatible with PI control. [1]

Limitations

- QSG quality strongly affects robustness.
- Sensitive to distortion unless bandwidth is reduced or enhanced QSG (e.g., SOGI) is used. [2,5]

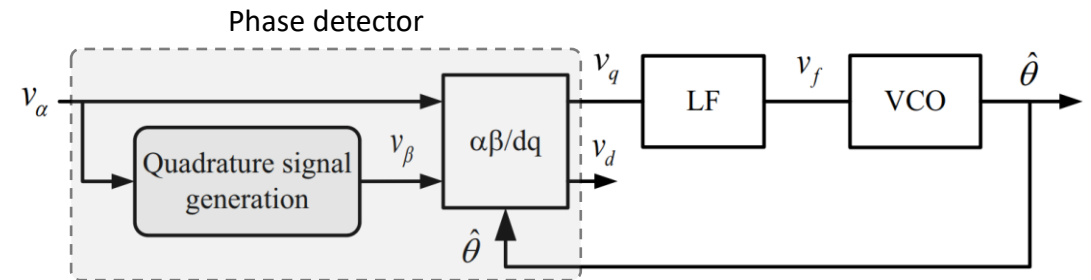


Figure 1.16. SRF-PLL for single-phase signals. [5]

T/4 transport delay-PLL

SRF-PLL methods for single phase systems

Concept [5,11]

- The T/4 transport-delay PLL reconstructs a quadrature signal by delaying the input voltage by T/4, corresponding to a 90° phase shift at the grid fundamental frequency.
- This creates pseudo-orthogonal pair ($v_\alpha = v_g$, $v_\beta = v_g(t - T/4)$) that enables the application of the $\alpha\beta \rightarrow dq$ transformation.

How it works [5,11]

- **QSG**: the delay block produces v_β , shifted by +90°.
- **dq transformation**: uses estimated angle θ' to compute v_q .
- **Loop filter (PI)**: regulates $\varepsilon \rightarrow$ estimated frequency ω' .
- **VCO/NCO**: integrates $\omega' \rightarrow$ updated θ' .

Strengths

- Very simple implementation — only requires a delay element and a standard SRF-PLL loop.
- Low computational burden.
- Good performance under clean single-phase sinusoidal conditions.

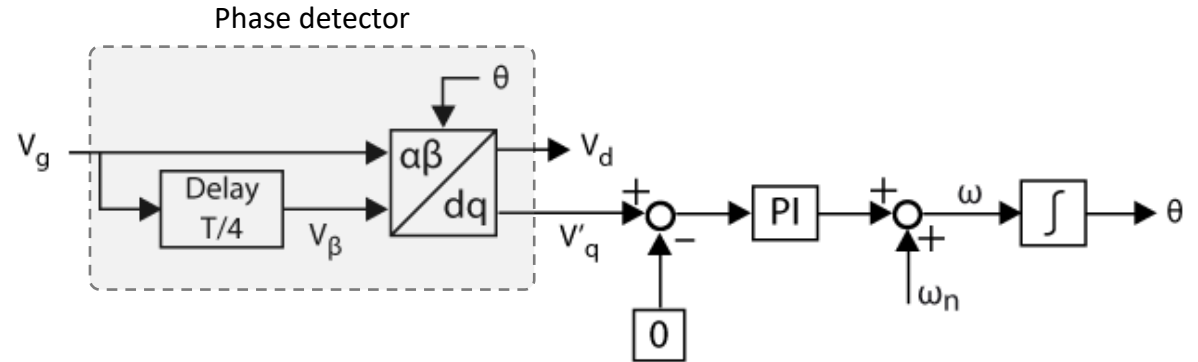


Figure 1.17. Block diagram of the T/4 transport delay-PLL. [11]

Limitations

- The T/4 phase shift is exact **only at the nominal fundamental frequency**, so performance degrades under frequency deviations.
- Highly sensitive to **harmonics and distortion**, since the delay cannot separate harmonic components.
- Not recommended for weak or distorted grids.

Inverse Park Transformation-PLL

SRF-PLL methods for single phase systems

Concept [5,11]

- The Inverse Park transformation-PLL generates a pseudo-orthogonal signal through a **closed-loop reconstruction** based on the dq frame itself.
- Instead of using a fixed delay, it computes (v_α, v_β) from the estimated dq components via the inverse Park transform, improving robustness to frequency variations compared with the T/4 method.

How it works [5,11]

- **Initial dq estimation:** the single-phase voltage V_g passes through a QSG block to obtain an initial pair.
- **Inverse Park transform:** uses the estimated angle θ' to regenerate $\alpha\beta$ components (v'_α, v'_β) .
- **LPF filtering:** filters out high-frequency components of the reconstructed signals.
- **Updated dq transform:** filtered $\alpha\beta$ voltages are transformed again to obtain refined v_d, v_q .
- **PLL loop:** $v_q \rightarrow$ phase error $\rightarrow$ PI $\rightarrow \omega' \rightarrow$ integrator $\rightarrow$ updated θ' .

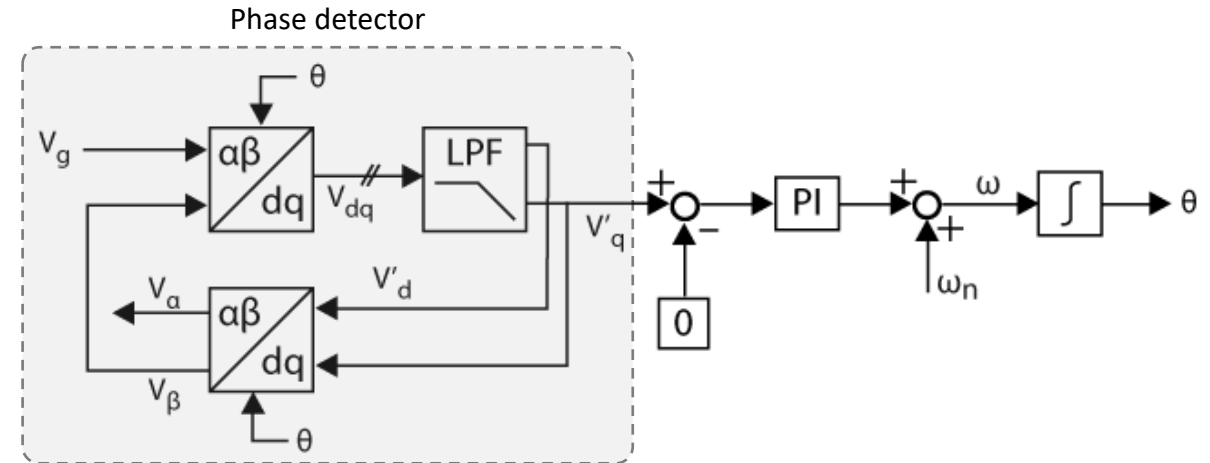


Figure 1.18. Block diagram of the inverse Park transformation-PLL. [11]

Strengths

- Better robustness than T/4-PLL under frequency drift, since the orthogonal signal is regenerated adaptively.
- Improved harmonic rejection thanks to the LPF applied to the reconstructed $\alpha\beta$ voltages.
- Still simple enough for embedded applications.

Limitations

- More computationally involved than T/4-PLL.
- Performance depends strongly on LPF tuning.
- Non-ideal under severe harmonic distortion [5].

Second Order Generalized Integrator (SOGI)

SRF-PLL methods for single phase systems

What Is a SOGI? [4,12]

- A Second-Order Generalized Integrator that generates:
 - A band-pass filtered in-phase component
 - A 90° phase-shifted quadrature component (QSG)
- Center frequency is tuned to ω' , enabling accurate $\alpha\beta$ orthogonal decomposition. [12]

Why It's Important

- Provides orthogonal signals without Park transform → ideal for single-phase or $\alpha\beta$ -domain synchronizers.
- Offers strong harmonic rejection and minimal phase error at the fundamental.
- Foundation for SOGI-PLL, SOGI-FLL, DSOGI-FLL, MSOGI-FLL. [2,4,5]

Key Parameters

- ω' (estimated frequency): centers the narrow band-pass.
- k (damping factor): governs filter bandwidth and transient response.
- Parameter design directly affects stability and tracking accuracy. [12]

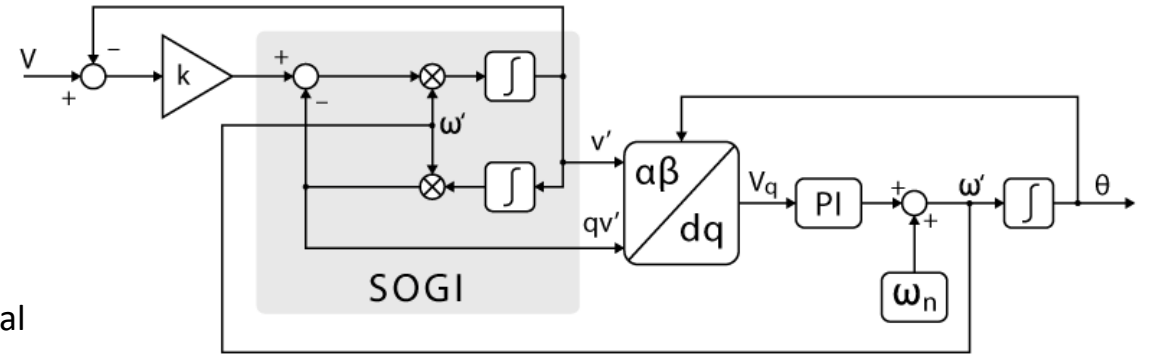


Figure 1.19. Block diagram of the SOGI PLL [12]

- The transfer function of the SOGI is given by:

$$\text{SOGI}(s) = \frac{v'}{k\varepsilon_v}(s) = \frac{\omega' s}{s^2 + \omega'^2}$$

- The two in-quadrature output signals of the adaptive filter in Figure 1.20, i.e., v' and qv' , are defined by the following transfer functions:

$$D(s) = \frac{v'}{v}(s) = \frac{k\omega' s}{s^2 + k\omega' s + \omega'^2}$$

$$Q(s) = \frac{qv'}{v}(s) = \frac{k\omega'^2}{s^2 + k\omega' s + \omega'^2}.$$

Second Order Generalized Integrator (SOGI)

SRF-PLL methods for single phase systems

- The two in-quadrature output signals of the adaptive filter in Figure 1.20, i.e., v' and qv' , are defined by the following transfer functions:

$$D(s) = \frac{v'}{v}(s) = \frac{k\omega' s}{s^2 + k\omega' s + \omega'^2}$$

$$Q(s) = \frac{qv'}{v}(s) = \frac{k\omega'^2}{s^2 + k\omega' s + \omega'^2}.$$

where ω' represents the estimated fundamental frequency and k the damping factor of the SOGI block.

These transfer functions are displayed below in the frequency domain to illustrate the impact of these parameters (here with $\omega' = 50\text{Hz}$).

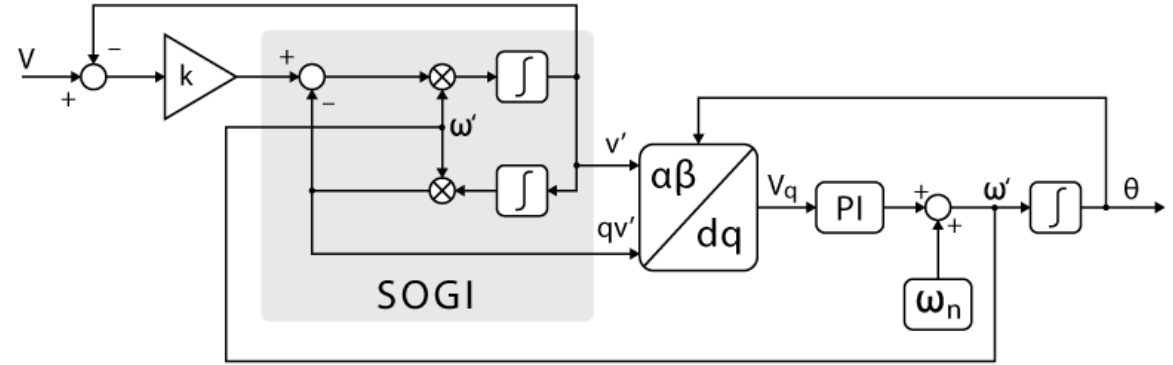


Figure 1.19. Block diagram of the SOGI PLL [12]

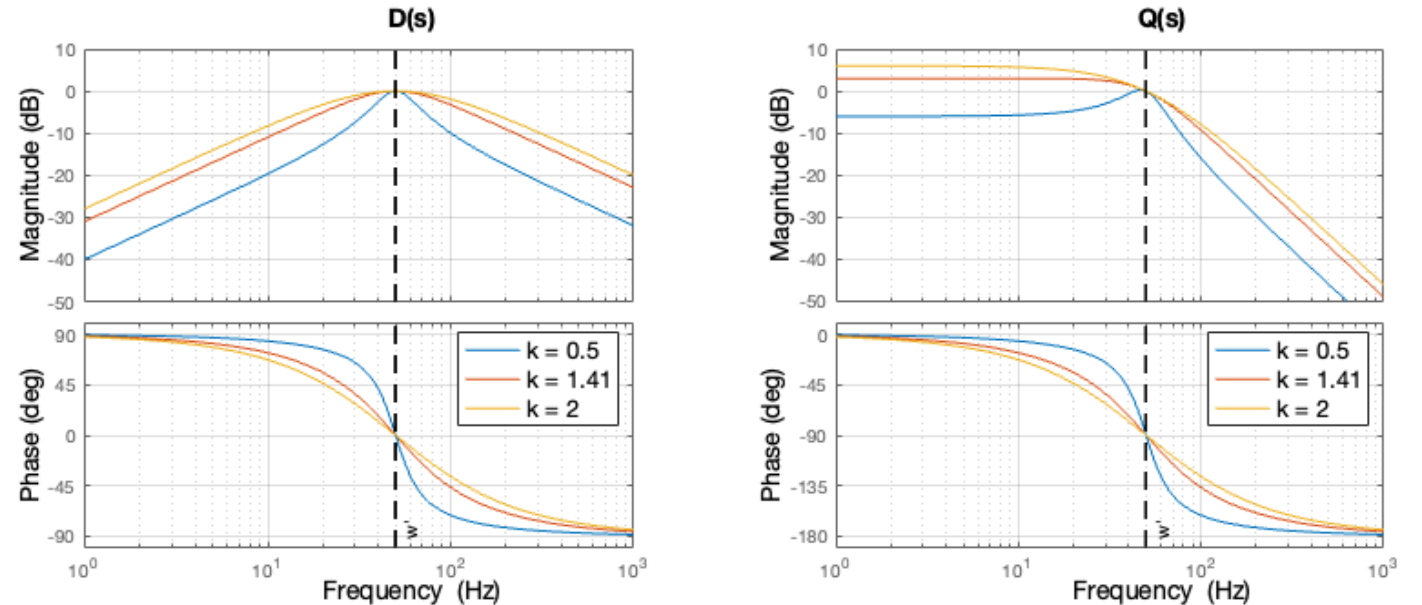


Figure 1.20. SOGI frequency domain behavior [12]

Second Order Generalized Integrator-PLL (SOGI-PLL)

SRF-PLL methods for single phase systems

Second-Order Generalized Integrator PLL (SOGI-PLL) [4,12]

- Uses SOGI QSG to create orthogonal signals, then a PLL (typically SRF-like) to lock θ ; better harmonic immunity than raw SRF-PLL, especially in single-phase.
- PLL operates in $\alpha\beta$, not in dq , which improves filtering and reduces sensitivity to grid distortion.
- Better performance than SRF-PLL in distorted but balanced grids due to band-pass nature of SOGI.
- Widely used in practice; tuning and digital implementation are well-documented.

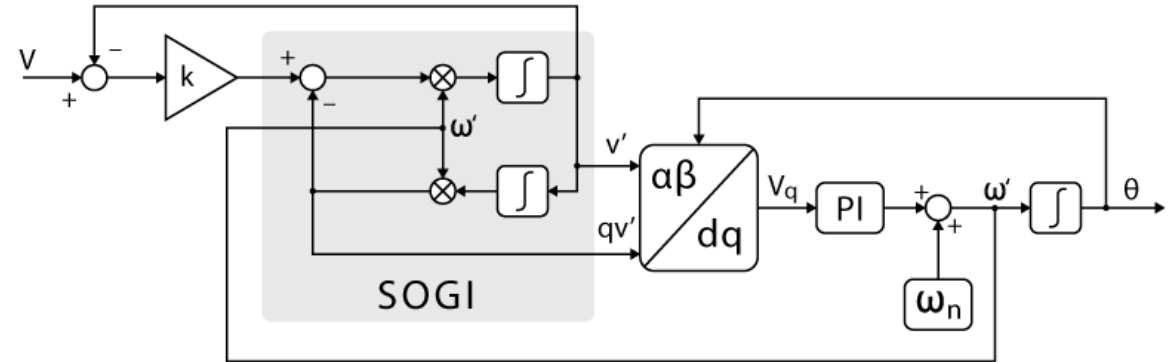


Figure 1.19. Block diagram of the SOGI PLL [12]

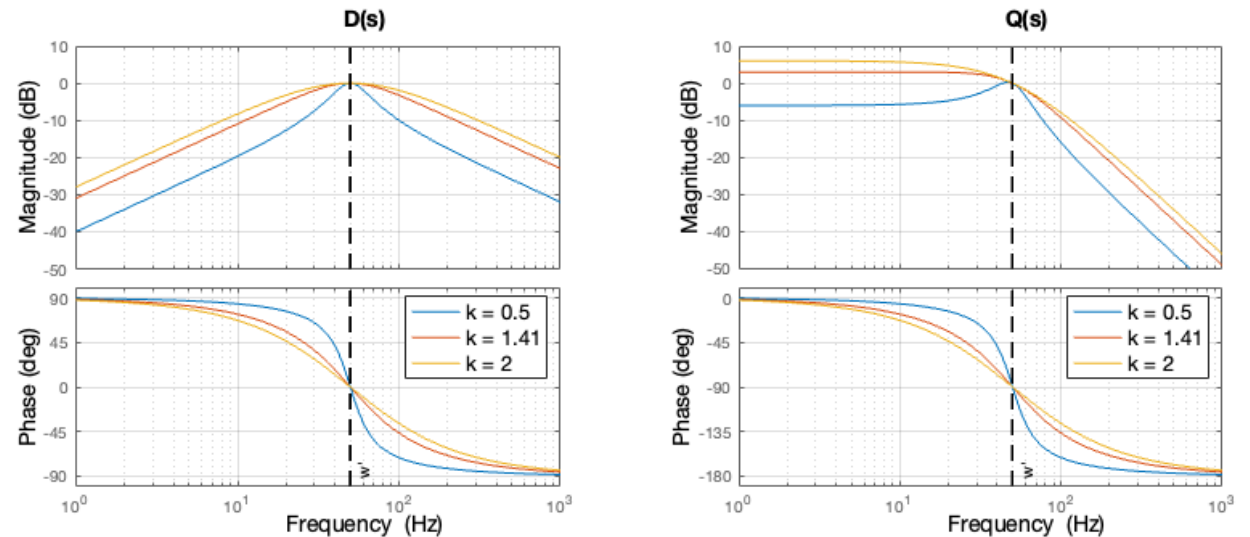


Figure 1.20. SOGI frequency domain behavior. [12]

Enhanced PLL (EPLL)

PLL methods for single phase systems

EPLL (Enhanced/Adaptive PLL) [5,13]

- Adaptive PLL using a nonlinear, self-tuning loop that estimates amplitude, phase and frequency simultaneously.
- Excellent rejection of harmonics, DC offsets, and frequency drifts, outperforming classical SRF-PLL in polluted grids.
- Provides smoother phase tracking under non-ideal, distorted waveforms due to adaptive gain and amplitude-normalized error.
- Suited for single-phase or heavily distorted systems where conventional PLLs suffer from ripple in the error signal.

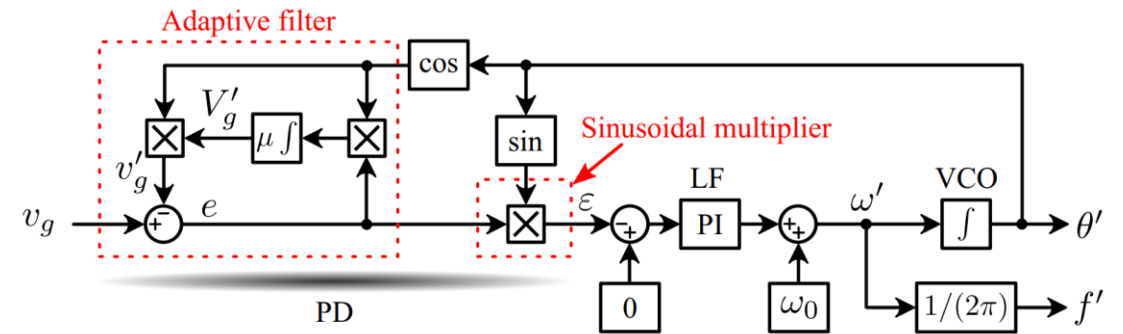


Figure 1.21. Block diagram of Enhanced phase locked loop (EPLL) system based on an adaptive filtering technique [13].

1.3: Grid Synchronization

Three Phase PLL



General structure of PLL for three phase systems

Concept [1,5,11]

- Three-phase PLLs benefit from the natural availability of orthogonal $\alpha\beta$ components, enabling direct application of the Clarke–Park transforms.
- The most commonly used type of PLL for these systems are synchronous reference frame-PLL (SRF-PLL) and SRF-PLL based modifications.
- These improved modifications offer enhanced robustness against unbalanced and/or harmonic distortion grid conditions.

Functional structure [11]

- **Stationary reference frame ($\alpha\beta$)**
 - Optional harmonic filtering using SOGI/DSOGI structures.
 - Optional positive-sequence extraction to improve robustness in unbalanced grids.
 - Provides clean v_α, v_β to the dq transformation.
- **Synchronous reference frame (dq)**
 - **Clarke–Park** transform produces v_d, v_q .
 - **Phase detector:** $\varepsilon = v_q$.
 - **Loop filter (PI):** regulates ε to generate ω' .
 - **NCO/VCO:** integrates ω' to obtain the updated phase θ' .

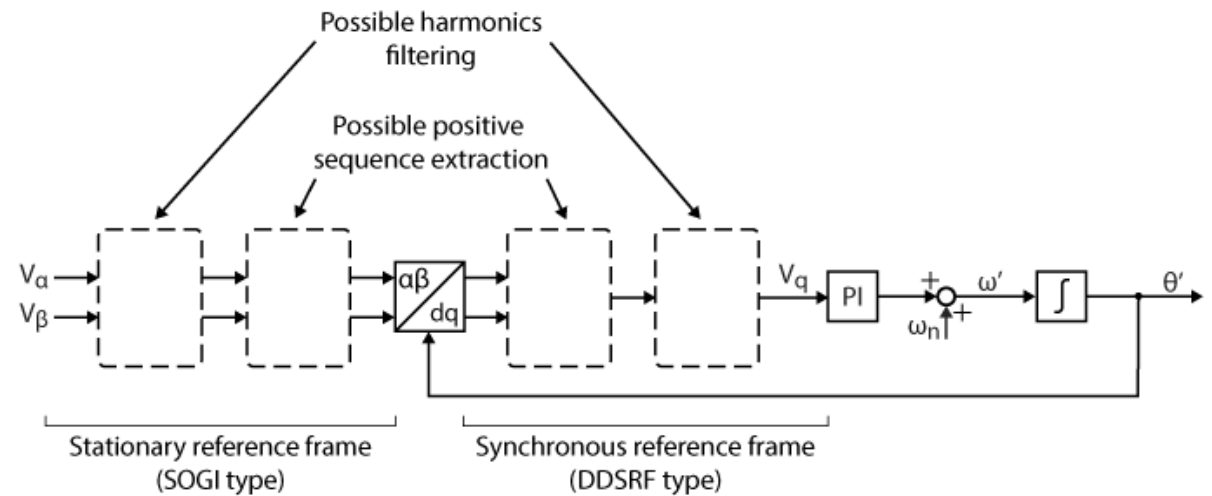


Figure 1.22. General three phase SRF-PLL and modifications [11]

Synchronous Reference Frame PLL (SRF-PLL)

PLL methods for three phase systems

Concept [1,14]

- The standard SRF-PLL for three-phase systems uses the Clarke–Park transformations, leveraging naturally available $\alpha\beta$ orthogonal voltages.
- Aligning the dq frame with the grid voltage forces $v_q \rightarrow 0$.

How it works [1,14]

- **$abc \rightarrow dq$ transformation:** Uses grid voltage v_{abc} and estimated angle θ' .
- **Phase detector:** $\varepsilon = v_q$.
- **Loop filter (PI):** Regulates ε to produce ω' .
- **Voltage Controlled Oscillator:** Integrates ω' to update θ' .

Strengths

- Very effective under balanced sinusoidal grids.
- Simple implementation; compatible with PI control. [1]

Limitations

- Under unbalanced grid:
 - Double-frequency ripple in $v_d/v_q \rightarrow$ phase jitter. [2]
 - Inaccurate amplitude/angle when V^- is present.
- Under distorted grid:
 - Harmonics propagate into angle estimation unless bandwidth is reduced. [2,5,11]

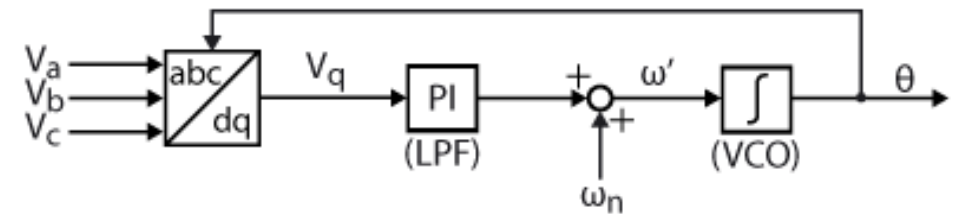


Figure 1.23. General principle of the three phase SRF PLL [14]

SRF-PLL: performance and limits

PLL methods for three phase systems

Performance Factors

- **Bandwidth:** Higher = faster tracking, but worse harmonic/noise rejection. Lower = stable under distortion but slower. [1,2]
- **Grid strength (SCR):** PLL dynamics may destabilize VSCs in weak grids; negative damping interactions observed. [3]
- **Filtering:** Essential to mitigate 2ω ripple and harmonics (LPF, notch, MAF, SOGI). [1,5]

Known Limits

- SRF-PLL cannot accurately extract V^+ (positive sequence of V_{abc}) under strong unbalance or deep voltage dips. [2]
- Harmonic distortion requires bandwidth reduction $\rightarrow$ degraded transient response. [5]
- In weak grids, PLL-based synchronization may excite resonances with current controllers. [3]

When PLL is Insufficient

- Deep asymmetrical faults, highly distorted networks, PSC/GFM behavior (where PLL may be bypassed). [3,6]

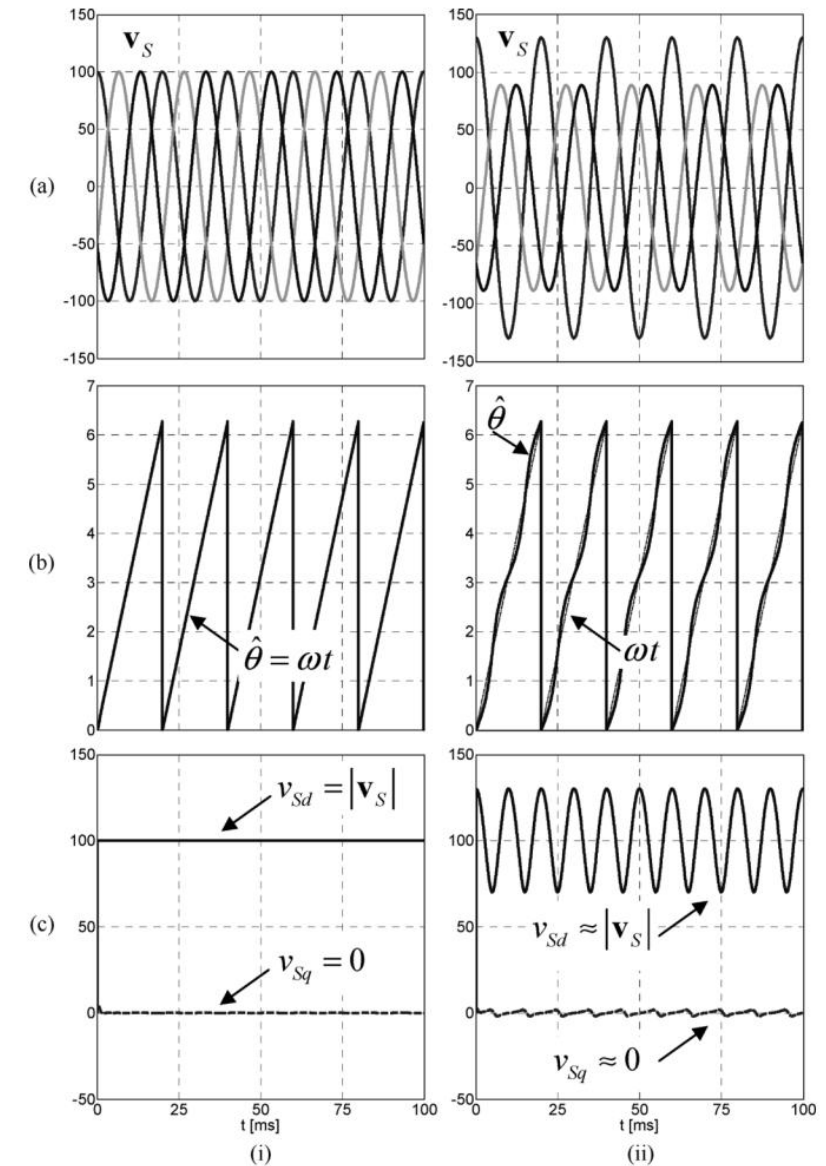


Figure 1.24. Response of the SRF-PLL with high bandwidth under i) balanced and ii) unbalanced conditions. (a) Utility voltage [V], (b) Detected phase angle [rad], and (c) SRF axes voltage [V]. [2]

Double Decoupled Synchronous Reference Frame (DDSRF-PLL)

PLL methods for three phase systems

Concept & Decoupling [2,14]

- The DDSRF-PLL improves the classical SRF-PLL by using two synchronous rotating reference frames:
 - One rotating at $+\omega \rightarrow$ extracts positive-sequence
 - One rotating at $-\omega \rightarrow$ extracts negative-sequence
- By isolating both sequences before entering the PLL loop, the 2ω characteristic ripple caused by negative-sequence disappears.
- This results in clean dq signals, perfect for accurate phase estimation during deep unbalance, dips and asymmetrical faults.

Mathematical model of decoupling [14]

- In the presence of unbalanced voltage, both positive and negative sequence components exist, and phase voltages can be expressed as in the dq reference frame as follow:

$$V_{dq^+} = V^+ \begin{bmatrix} \cos(\omega t - \theta') \\ \sin(\omega t - \theta') \end{bmatrix} + V^- \begin{bmatrix} \cos(-\omega t - \theta' + \phi^-) \\ \sin(-\omega t - \theta' + \phi^-) \end{bmatrix}$$

$$V_{dq^-} = V^+ \begin{bmatrix} \cos(\omega t + \theta') \\ \sin(\omega t + \theta') \end{bmatrix} + V^- \begin{bmatrix} \cos(-\omega t + \theta' + \phi^-) \\ \sin(-\omega t + \theta' + \phi^-) \end{bmatrix}$$

- With $V^\pm$ the positive and negative sequence voltage, ωt the real grid phase, θ' the estimated grid phase and ϕ^- the negative sequence phase offset.

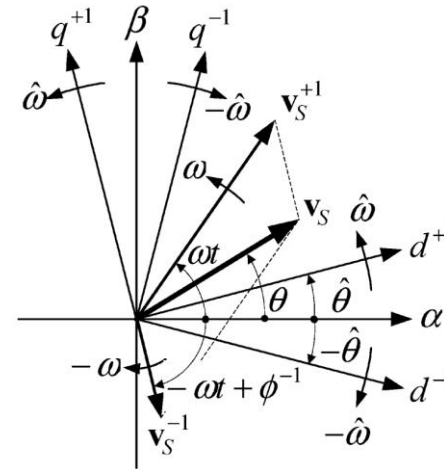


Figure 1.25. Voltage vectors and axes of the double synchronous reference frame (DSRF). [2]

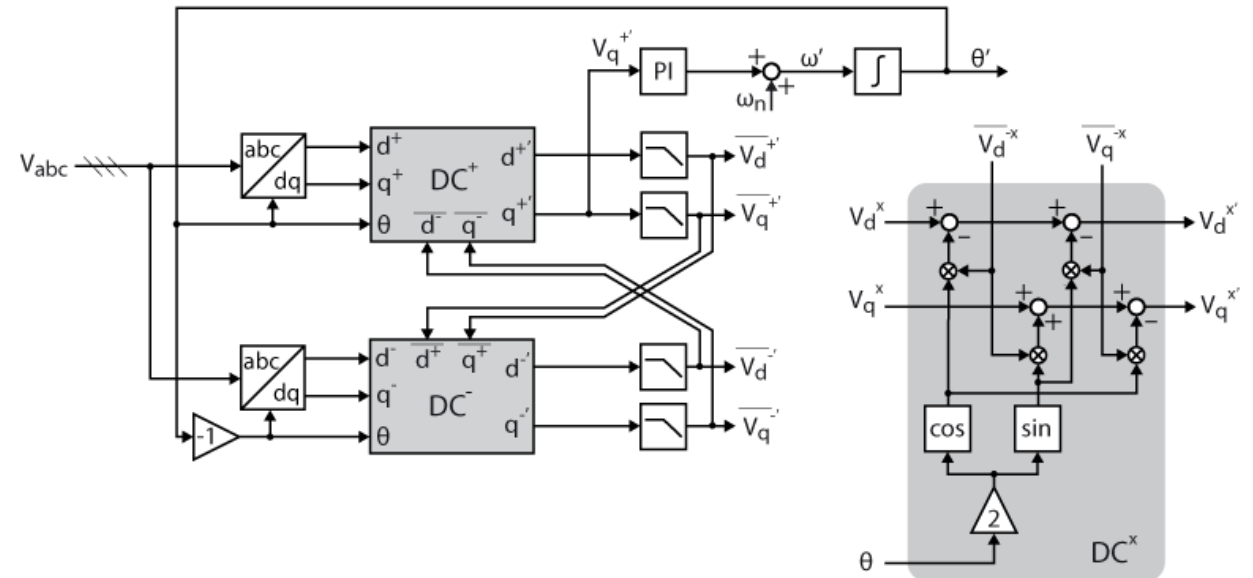


Figure 1.26. Block diagram of the DDSRF-PLL. [2,14]

Double Decoupled Synchronous Reference Frame (DDSRF-PLL)

PLL methods for three phase systems

Mathematical model of decoupling (continue) [14]

- Assuming a reasonably well-parameterized PLL is used, it can be assumed that $\omega t \approx \theta'$ resulting in the following voltages:

$$V_{dq+} \approx V^+ \begin{bmatrix} 1 \\ 0 \end{bmatrix} + V^- \begin{bmatrix} \cos(-2\omega t + \phi^-) \\ \sin(-2\omega t + \phi^-) \end{bmatrix} ; \quad V_{dq-} \approx V^+ \begin{bmatrix} \cos(2\omega t) \\ \sin(2\omega t) \end{bmatrix} + V^- \begin{bmatrix} \cos(\phi^-) \\ \sin(\phi^-) \end{bmatrix}$$

- Indeed, the positive and negative sequences are coupled.
- To synchronize the PLL on V_q^+ the oscillating term at 2ω must be removed. This is achieved by reconstructing the perturbation signal using trigonometry and filtering the negative sequence voltage.
- The reconstructed signal is then subtracted from the positive sequence voltage to obtain a pure positive sequence voltage.
- The reconstruction of the parasitic signal is:

$$V^- \begin{bmatrix} \cos(-2\omega t + \phi^-) \\ \sin(-2\omega t + \phi^-) \end{bmatrix} \stackrel{\theta' \approx \omega t}{=} \begin{bmatrix} \overline{v_{d-}'} \\ \overline{v_{q-}'} \end{bmatrix} \begin{bmatrix} \overline{V^- \cos(\phi^-)} \cos(2\theta') + \overline{V^- \sin(\phi^-)} \sin(2\theta') \\ -\overline{V^- \cos(\phi^-)} \sin(2\theta') + \overline{V^- \sin(\phi^-)} \cos(2\theta') \end{bmatrix}$$

- The above v_{dq-}' components are obtained by filtering the negative sequence voltage with a low-pass filter (LPF) whose cutoff frequency is $\leq 2\omega$. It is recommended to use a first-order Butterworth filter with a cutoff frequency of $\omega_0 = \omega/\sqrt{2}$.

$$LPF(V_{dq-}) = \overline{v_{dq-}'} \approx V^- \begin{bmatrix} \cos(\phi^-) \\ \sin(\phi^-) \end{bmatrix}$$

- Finally, the reconstructed signal is subtracted from the perturbed signal, allowing for effective decoupling of the sequences:

$$V_{dq+} \approx V^+ \begin{bmatrix} 1 \\ 0 \end{bmatrix} + V^- \begin{bmatrix} \cos(-2\omega t + \phi^-) \\ \sin(-2\omega t + \phi^-) \end{bmatrix} - V^- \begin{bmatrix} \cos(-2\theta' + \phi^-) \\ \sin(-2\theta' + \phi^-) \end{bmatrix} \approx V^+ \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Double Decoupled Synchronous Reference Frame (DDSRF-PLL)

PLL methods for three phase systems

Complete DDSRF-PLL Structure [2,14]

- **Clarke transform:** Converts $v_{abc} \rightarrow v_{\alpha}, v_{\beta}$
- **Two Park transforms:**
 - $+\omega \rightarrow v_d^+, v_q^+$
 - $-\omega \rightarrow v_d^-, v_q^-$
- **Decoupling Cell:** Removes all 2ω cross-coupling, giving:
 $V_q^{+,dec} \approx$ pure positive sequence for PLL
- **PLL Loop:** error $\rightarrow$ loop filter (PI) $\rightarrow$ VCO (integral)

Characteristics

- Eliminates 2ω ripple completely $\rightarrow$ zero phase jitter even in severe unbalance.
- Fast dynamic performance during asymmetrical faults, sags, and dips.
- Highly accurate positive-sequence extraction, even when V^- / V^+ is large.
- Industry standard for wind, solar, STATCOM, and grid-code LVRT/FRT compliance.
- More robust than SRF-PLL but slightly more computationally intensive.

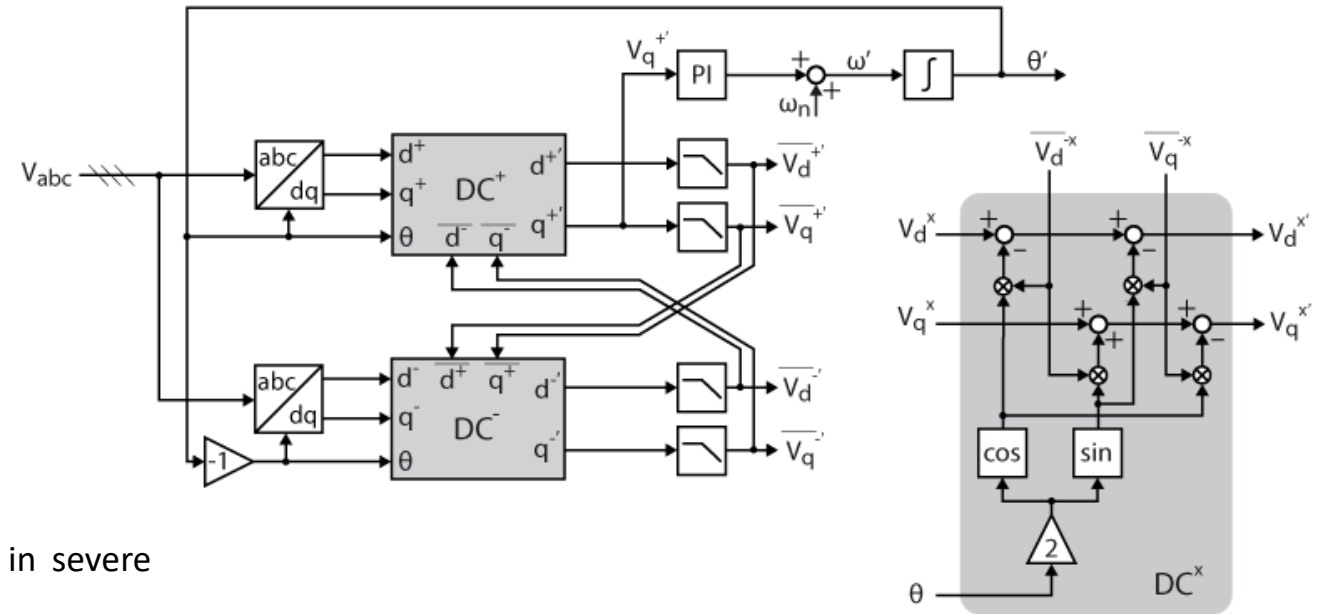


Figure 1.26. Block diagram of the DDSRF-PLL. [2,14]

Dual Second-Order Generalized Integrator (DSOGI-PLL)

PLL methods for three phase systems

DSOGI-PLL [12,15]

- DSOGI-PLL extends the SOGI-PLL concept to the three-phase $\alpha\beta$ frame using two SOGIs (for α and β channels) to generate orthogonal signals and accurate 90°-shifted versions.
- These orthogonal signals feed a Positive-Sequence Calculator (PSC) implemented using instantaneous symmetrical components in $\alpha\beta$.
- The resulting positive-sequence voltage is then processed by an SRF-PLL, which provides frequency adaptation and ensures stable phase tracking under grid disturbances.

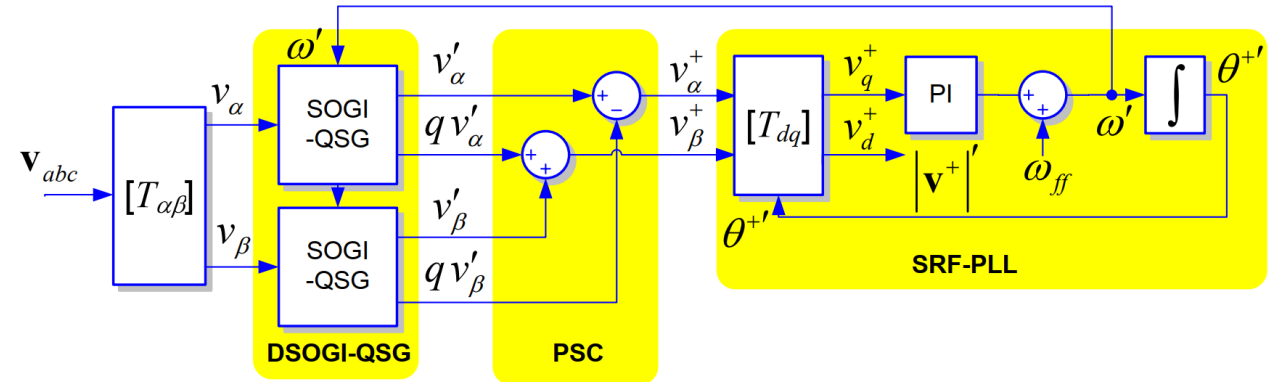


Figure 1.27. Block diagram of the DSOGI-PLL. [15]

Strengths

- Excellent positive-sequence extraction under deep unbalance and voltage sags.
- Built-in harmonic rejection from the SOGI stages (acts as band-pass for v^+ and notch for v^-).
- Fast adaptation to frequency deviations; robust in weak and distorted grids.
- Clean and stable $v_d^+, v_q^+ \rightarrow$ high-quality PLL error signal.

Limitations

- More computationally intensive than single-SOGI or basic SRF-PLL.
- Performance depends on tuning of SOGI parameters and PI gains of the SRF-PLL.
- Slightly slower than raw SRF-PLL due to preprocessing, but vastly more robust.

Multi Second-Order Generalized Integrator (MSOGI-PLL)

PLL methods for three phase systems

MSOGI-PLL = PLL fed by a multi-SOGI (MSOGI) front-end (parallel SOGIs **tuned at selected harmonics**) to improve harmonic rejection before the PLL. (This is the PLL counterpart of MSOGI-FLL). [4, 12]

Concept

- Multiple SOGIs tuned at $1\omega, 5\omega, 7\omega, \dots$ build a harmonic decoupling network (HDN). Their outputs are recombined to notch out targeted harmonics from the fundamental template that drives the PLL.

Benefit

- Reduces harmonic leakage into $v_q \rightarrow$ cleaner PLL error and reduced bandwidth compromises under distortion. Complements DDSRF (for sequence) with MSOGI (for harmonics).

When to use

- Distorted but mostly balanced grids where harmonic content (e.g., 5th/7th/11th) dominates over unbalance.

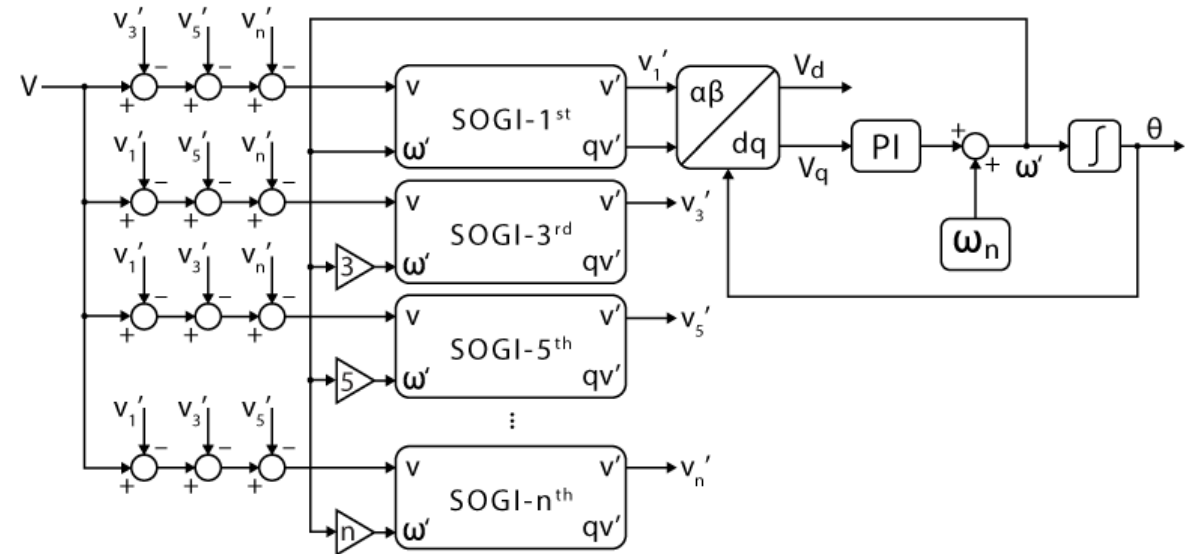


Figure 1.28. Block diagram of the MSOGI-PLL. [12]

Comparison of the three-phase PLLs: SRF, DDSRF, MSOGI

PLL methods for three phase systems

- In [11] three-phase grid synchronization methods SRF-PLL, DDSRF-PLL, MSOGI-PLL are compared below to observe their responses under disturbed conditions. For benchmarking purposes, disturbances will include i) a 60° phase jump, ii) a 2 Hz frequency jump, iii) 5th harmonic distortion with an amplitude of 0.1 p.u., and iv) a phase-to-phase fault in the grid.

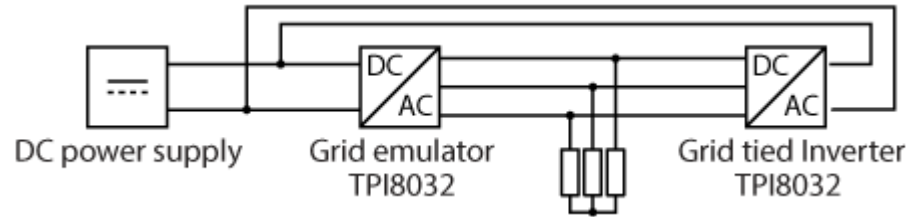


Figure 1.29. Electrical diagram of the setup. [11]

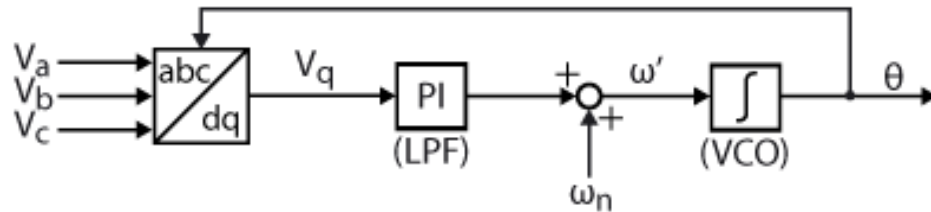


Figure 1.23. General principle of the **SRF PLL** [14]

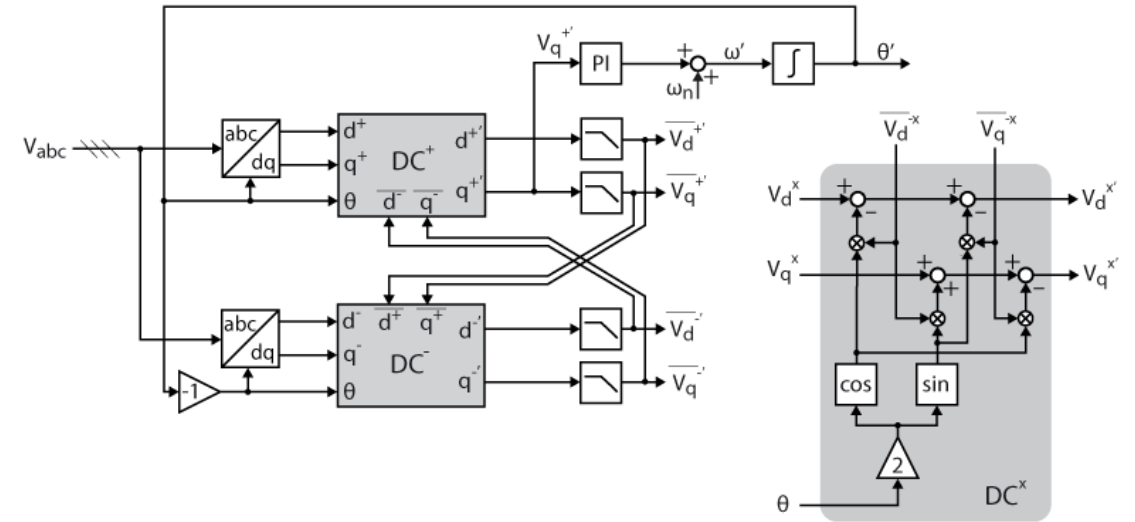


Figure 1.26. Block diagram of the **DDSRF-PLL**. [2,14]

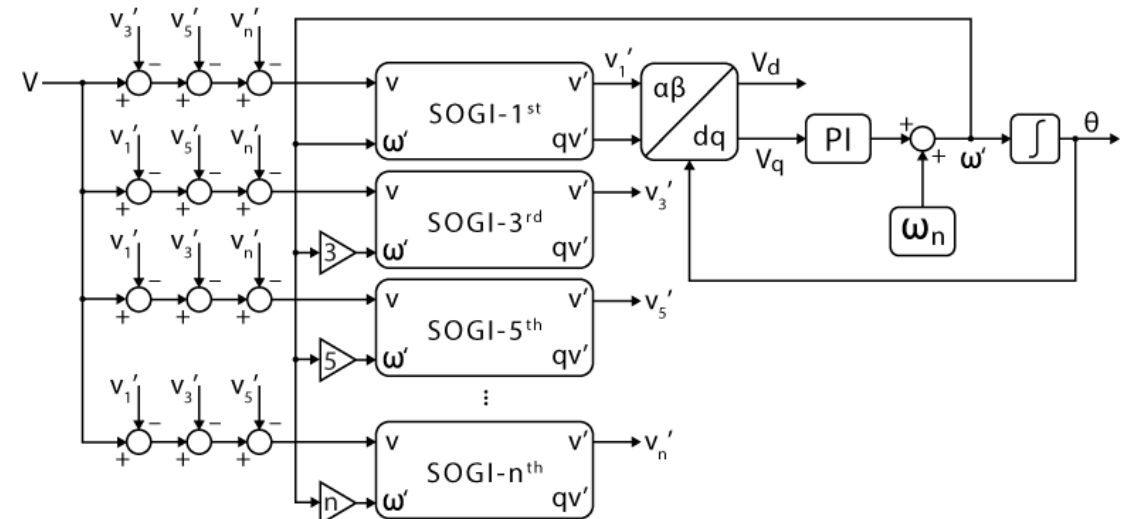


Figure 1.28. **MSOGI-PLL** block diagram. [12]

Comparison of the three-phase PLLs: SRF, DDSRF, MSOGI

PLL methods for three phase systems

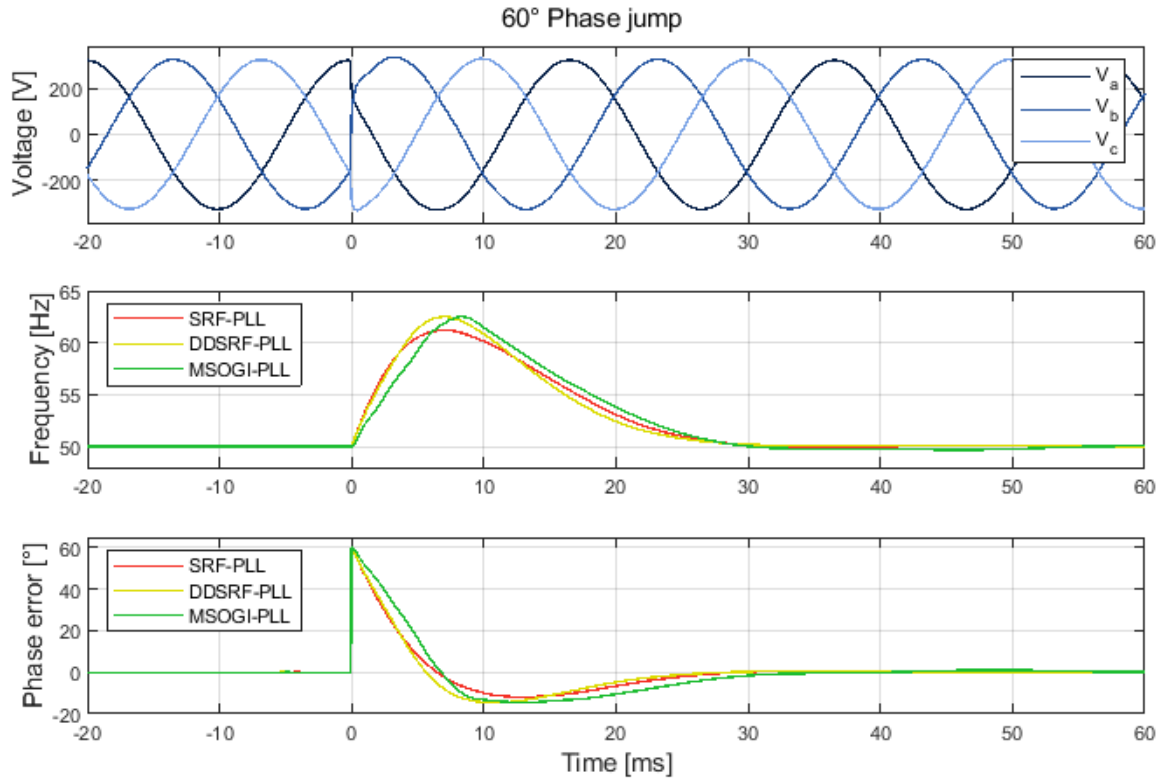


Figure 1.30. Responses to a 60° phase step. [11]

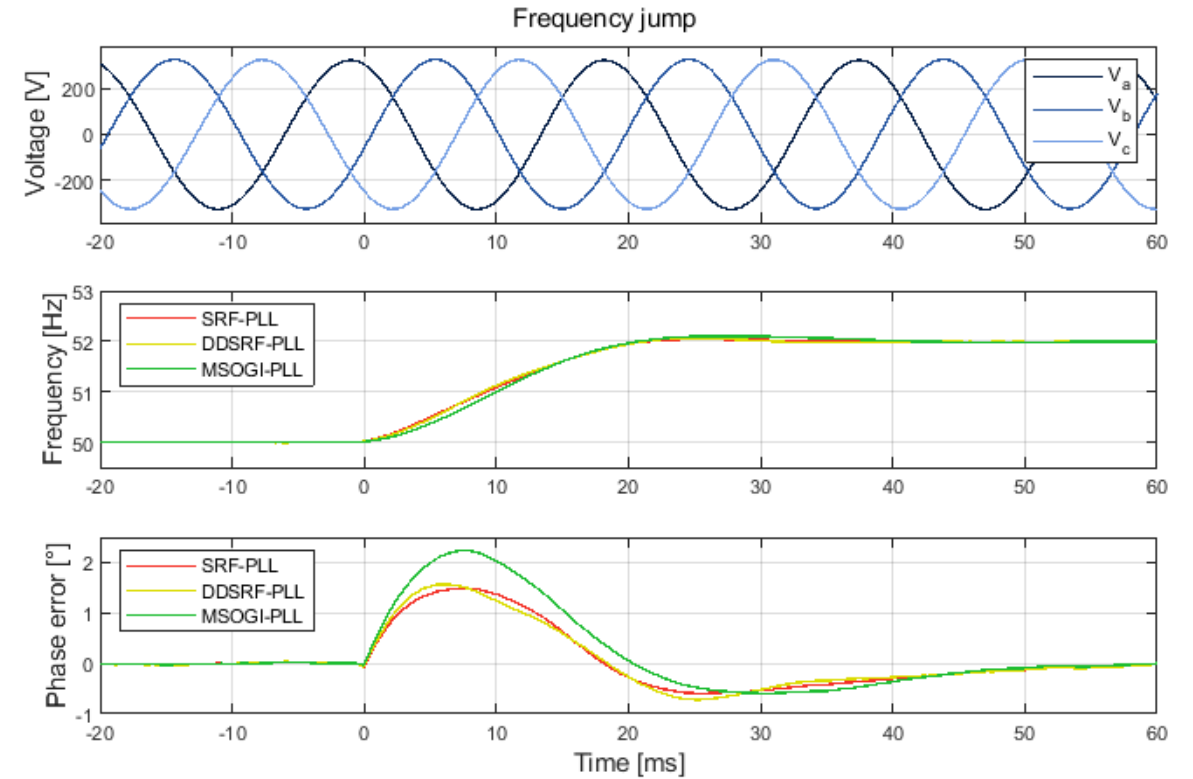


Figure 1.31. Responses to a 2Hz frequency step. [11]

Comparison of the three-phase PLLs: SRF, DDSRF, MSOGI

PLL methods for three phase systems

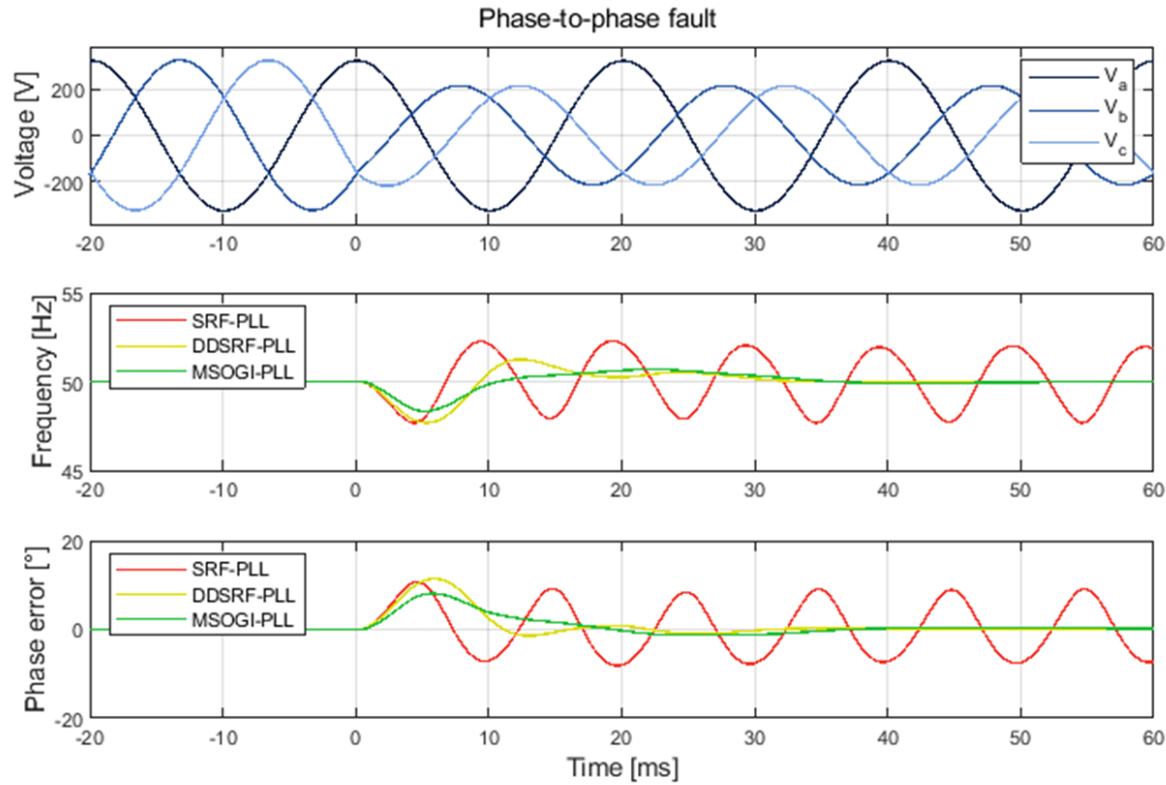


Figure 1.32. Responses to unbalanced conditions. [11]

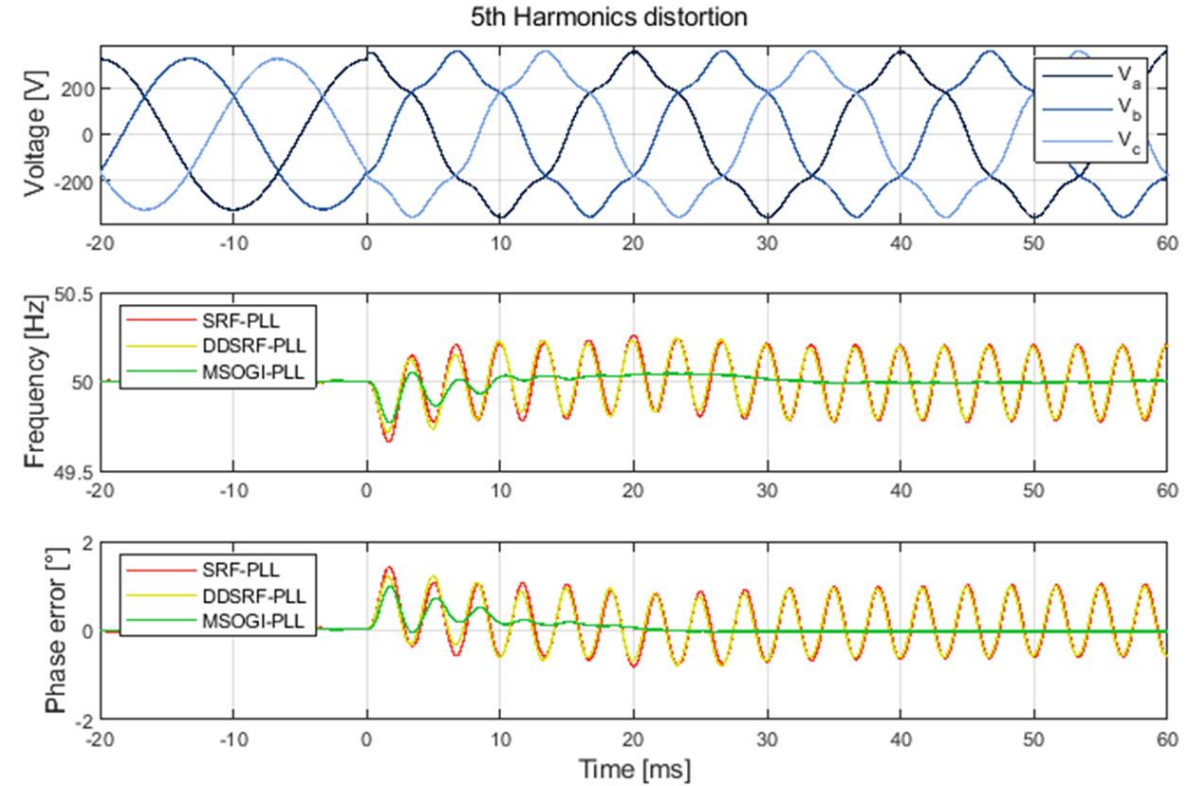


Figure 1.33. Responses to 5th harmonics distortion. [11]

1.4: Grid Synchronization

Frequency Locked Loop (FLL)



SOGI Frequency Locked Loop (SOGI-FLL) [4]

FLL methods for single phase systems

Definition

- A Frequency-Locked Loop (FLL) estimates grid frequency using frequency-error adaptation rather than phase-error minimization.
- The SOGI-FLL combines a Second-Order Generalized Integrator (SOGI) with an adaptive FLL law to obtain both orthogonal signals and an accurate frequency estimate.
- Operates entirely in the $\alpha\beta$ domain (no Park transform required).

How It Works

- The SOGI generates in-phase v' and quadrature qv' components.
- The FLL computes a frequency-error signal using the product of quadrature and error components.
- The estimated frequency ω' adapts online and is fed back to the SOGI resonance frequency.
- Angle θ' can be recovered by integrating ω' if needed.

Advantages

- Naturally robust to frequency drift and grid variations.
- Useful under distortion where PLL bandwidth must be severely reduced.
- Commonly used as the frequency-estimation block inside SOGI-PLL or MSOGI-FLL structures.

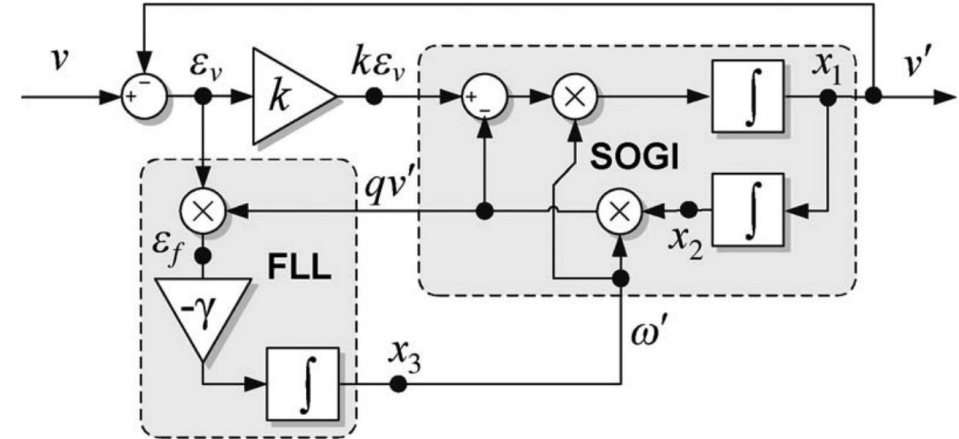


Figure 1.34. SOGI-FLL, a single-phase grid synchronization system. [4]

Limitations

- Slower than PLLs for phase tracking.
- Requires careful tuning of SOGI gain k and FLL gain γ .
- May require amplitude normalization for improved stability.

Multi SOGI Frequency Locked Loop (MSOGI-FLL) [4]

FLL methods for single phase systems

Definition

- Uses parallel SOGIs tuned at fundamental and selected harmonics.
- Provides positive-/negative-sequence extraction and multi-harmonic components.
- Intended for heavily distorted single-phase grids..

How It Works

- Each SOGI-QSG tracks one frequency (fundamental, 3rd, 5th, etc.).
- A common FLL updates the fundamental frequency and tunes all SOGIs accordingly.
- The harmonic decoupling network (HDN) prevents cross-coupling among SOGIs and isolates each harmonic.

Advantages

- Excellent harmonic immunity.
- Frequency-adaptive, useful in weak and distorted networks.
- Enables grid-connected inverters to implement multi-harmonic compensation or selective harmonic elimination.

Limitations

- Higher computational burden than single-SOGI systems.
- Requires careful tuning of SOGI bandwidth (k) and FLL gain (γ).
- More complex than DDSRF-PLL or SOGI-PLL for basic synchronization.

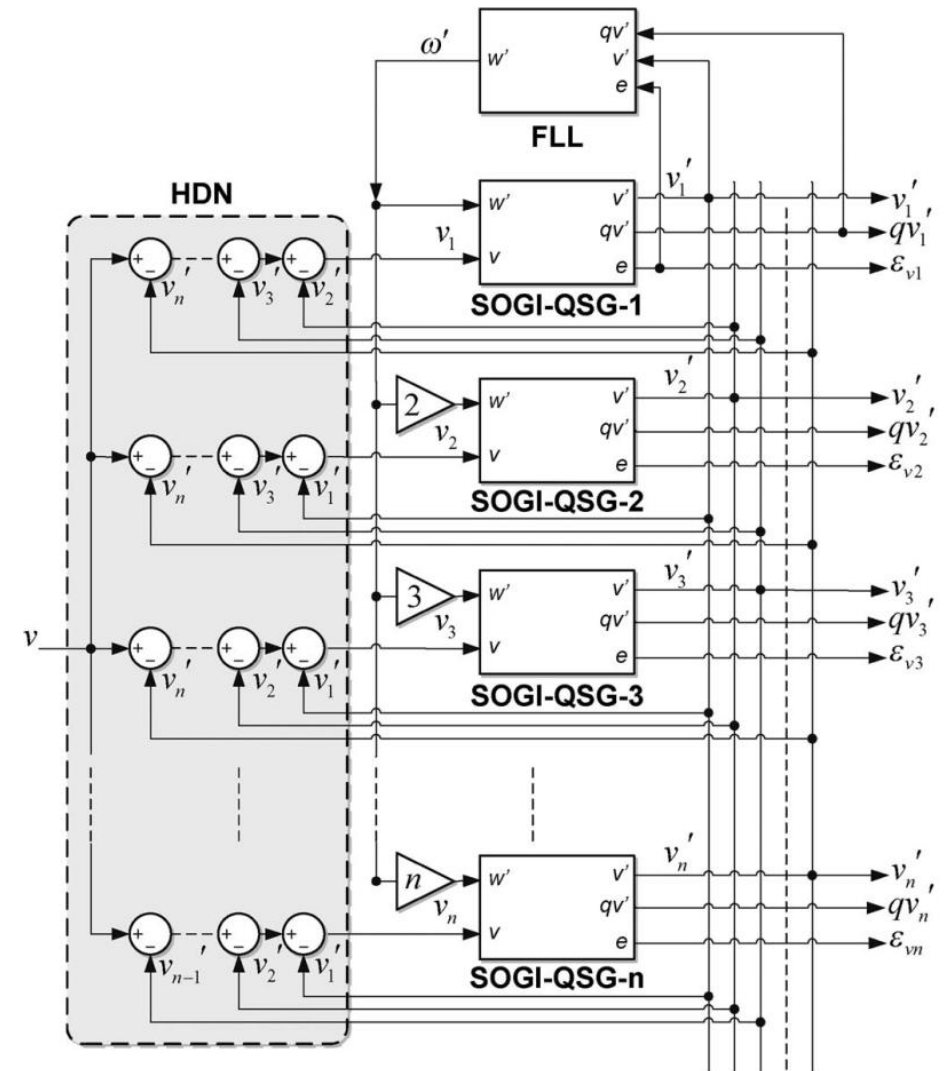


Figure 1.35. Block diagram of MSOGI-FLL. [4]

Dual SOGI Frequency Locked Loop (DSOGI-FLL) [4]

FLL methods for three phase systems

Definition

- Extends SOGI-FLL by applying two SOGIs (α and β channels) in the stationary frame.
- Produces orthogonal in-phase/quadrature signals, enabling frequency-adaptive and sequence-aware synchronization.

How It Works

- Each channel processed by a SOGI-QSG $\rightarrow v'_\alpha, qv'_\alpha, v'_\beta, qv'_\beta$.
- The FLL computes ω' with amplitude-normalized gain, enhancing robustness.
- Optional PNSC block yields positive/negative sequences for advanced control.

Advantages

- Very robust under grid unbalance.
- Better harmonic immunity than SRF-PLL alone.
- Provides accurate frequency for PLL hybrids or direct angle estimation.

Limitations

- Slower phase alignment compared to PLL-only systems.
- Still needs an angle estimation block for full synchronization.

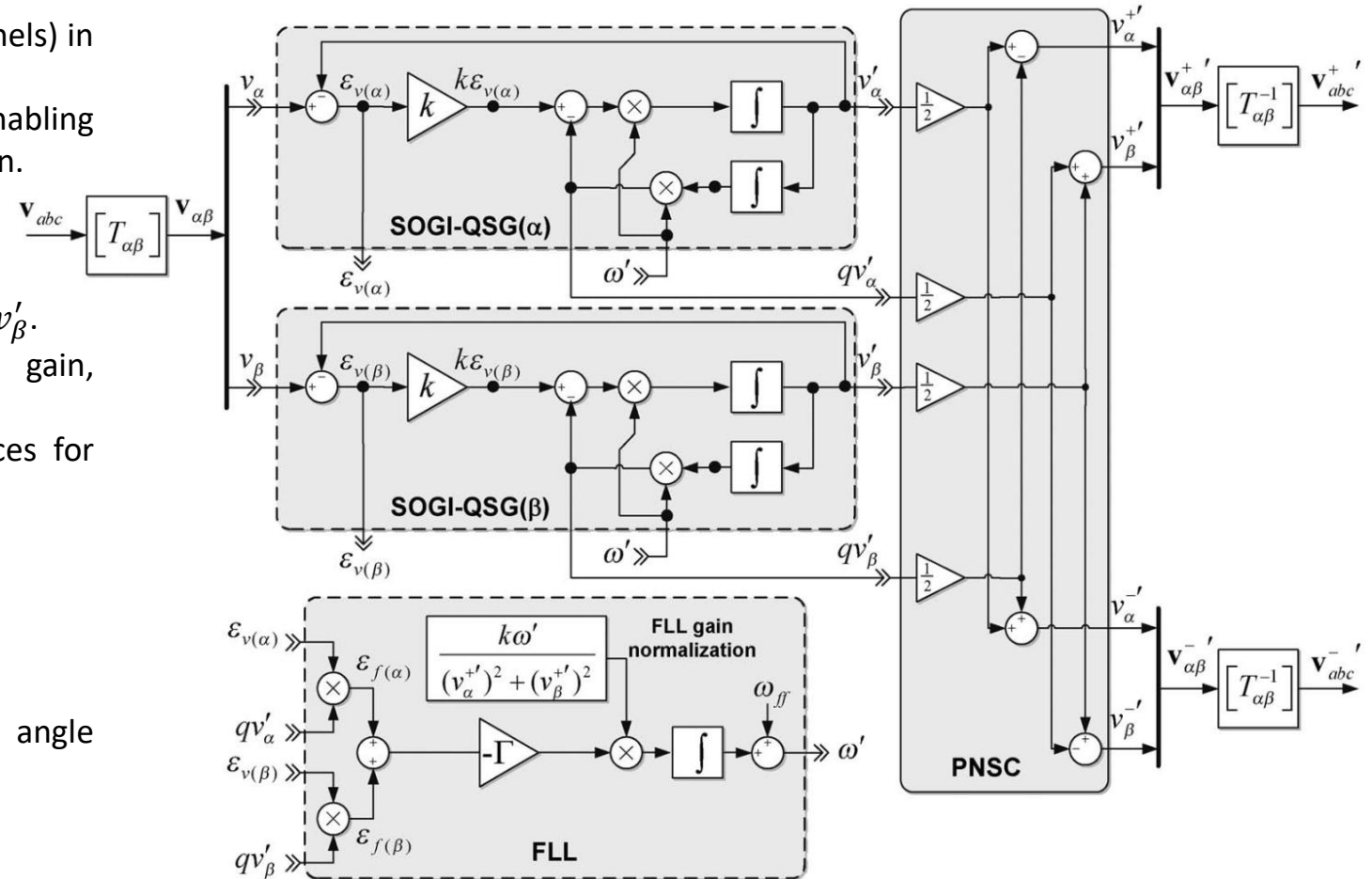


Figure 1.36. Block diagram of the DSOGI-FLL divided into its building blocks: SOGI-QSG(β), SOGI-QSG(α), FLL, and PNSC. [4]

Multi Dual SOGI Frequency Locked Loop (MDSOGI-FLL) [4]

FLL methods for three phase systems

Definition

- Multi-harmonic extension of DSOGI-FLL using multiple DSOGI-QSG units (1st, 5th, 7th, ...).
- Delivers harmonic-resolved $\alpha\beta$ signals plus positive/negative sequences for each harmonic.

How It Works

- For each harmonic order, two SOGIs generate α/β quadratures.
- PNSC block separates sequences per harmonic.
- A single FLL adapts the resonance frequency for all DSOGIs.

Advantages

- Exceptional robustness under heavily distorted and unbalanced three-phase grids.
- Supplies multi-harmonic information for advanced filtering or control.
- Fully frequency-adaptive.

Limitations

- Highest computational cost of all FLL methods.
- Complex tuning for multiple resonant channels.
- Typically used in advanced PQ or active filtering applications.

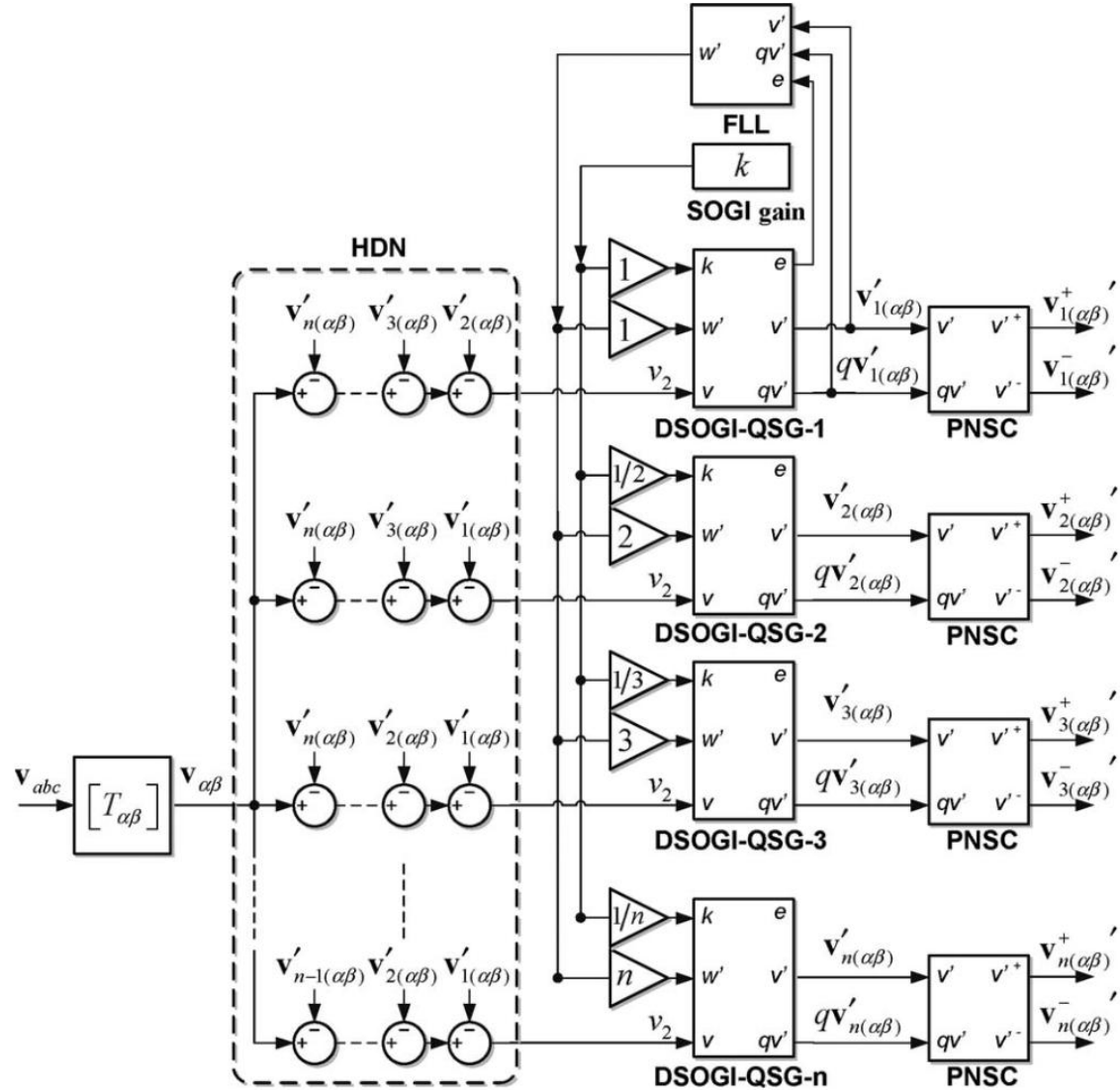


Figure 1.37. MSOGI-FLL for three-phase systems able to synchronize with “n” harmonic components of the input vector. [4]

1.5: Grid Synchronization

Summary of PLL methods



Comparative summary of PLL methods [1,2,4,5,10,12,13,14,15]

Table 1.1. Comparative summary single phase PLL synchronization methods.

Method	Harmonics immunity	Speed / Dynamics	Complexity	Typical use: 1-phase system
SRF-PLL (T/4 Transport Delay)	Low	Medium–High at nominal frequency	Low	Clean & stiff grids
SRF-PLL (Inverse Park Transform)	Low–Medium	Medium	Medium	With moderate frequency variations
SOGI-PLL	Medium–High (band-pass QSG)	Medium–High	Medium	Distorted single-phase grids
EPLL	High (adaptive filtering)	Medium	Medium	With moderate distortion and frequency drift

Table 1.2. Comparative summary three phase PLL synchronization methods.

Method	Unbalance tolerance	Harmonics immunity	Speed / Dynamics	Complexity	Typical use: 3-phase system
SRF-PLL	Low (2ω ripple)	Low–Medium (needs bandwidth $\downarrow$)	High on clean grid	Low	Clean & stiff grids
DDSRF-PLL	Very High (V^+/V^- decoupling)	Medium (add filtering if needed)	Medium	Medium–High	Unbalanced networks (faults, dips). Without harmonics
DSOGI-PLL	High	Medium–High (band-pass QSG)	Medium–High	Medium-High	Unbalanced and distorted grids
MSOGI-PLL	Very High	Very High (multi-resonant)	Medium	High	Distorted & unbalanced, multi-harmonic monitoring/control

Section 2: Operation Modes of Grid-Connected Converters

Power Electronic Control Fundamentals



Filter for Grid-Connected Converters

Why grid filters matter

- Reduce **switching ripple / EMI** at PCC and protect synchronizers from HF noise. [1,5,13]
- Avoid **control interaction**: resonance can couple with current loop & PLL/FLL; co-design bandwidths. [1,6,13]
- In general, three types of filters are implemented: L / LC / LCL

L filter: single inductor between inverter and grid.

- Pros: simple, robust; minimal resonance concerns.
- Cons: weaker HF attenuation $\rightarrow$ larger L, more voltage drop; slower dynamics.
- Use when: low power / relaxed EMI or as baseline topology.

LC filter: series L with shunt C to ground/grid.

- Pros: better attenuation than L; moderate component sizes.
- Cons: introduces resonance, so needs passive or active damping.
- Use when: moderate attenuation required; controller can handle added pole/zero.

LCL filter: inverter-side L, grid-side L, shunt C in between.

- Pros: high attenuation with compact parts; industry standard in MV/LV VSCs.
- Cons: sharp resonant peak; requires damping branch or active damping.
- Use when: stringent EMI limits and tight footprint; plan damping and sampling carefully.

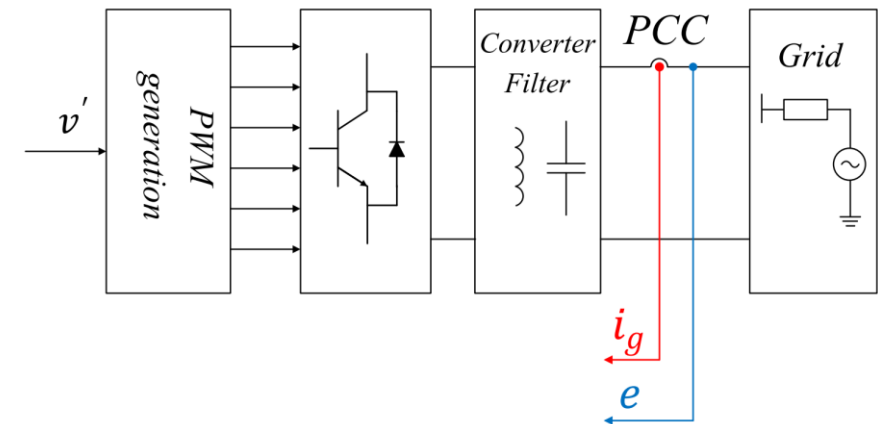


Figure 1.39. Simple grid-connected converter scheme. [6]

Operation Modes: Grid-Following / Grid-Supporting / Grid-Forming

Depending the connection type to grid and operation possibilities, the inverter can be classified in three categories:

Grid-Following (GFL)

- Most common for PV and classic inverters.
- Controlled as a current source: tracks i_d/i_q given by P/Q references.
- Requires a solid grid + PLL.
- Fast current loop; outer DC-link or power loop.

Grid-Supporting (GSM)

- Hybrid mode: GFL structure + voltage/frequency support via droop.
- Provides ancillary services (voltage regulation, inertia emulation).
- Usually uses nested loops: voltage outer loop for DC-Link control and current inner loop.

Grid-Forming (GFM)

- Controls voltage and frequency directly → acts like a virtual generator.
- Can include droop control (P-f, Q-V).
 - Grid Forming Power Droop Connected (**GFM-PD**)
- Can be synchronized with a Power Synchronization Loop (P- θ).
 - Grid Forming Power Synchronization Loop (**GFM-PSync**)
- Enables operation in islanded systems or microgrids.
 - Grid Forming Islanded Mode (**GFM-I**)

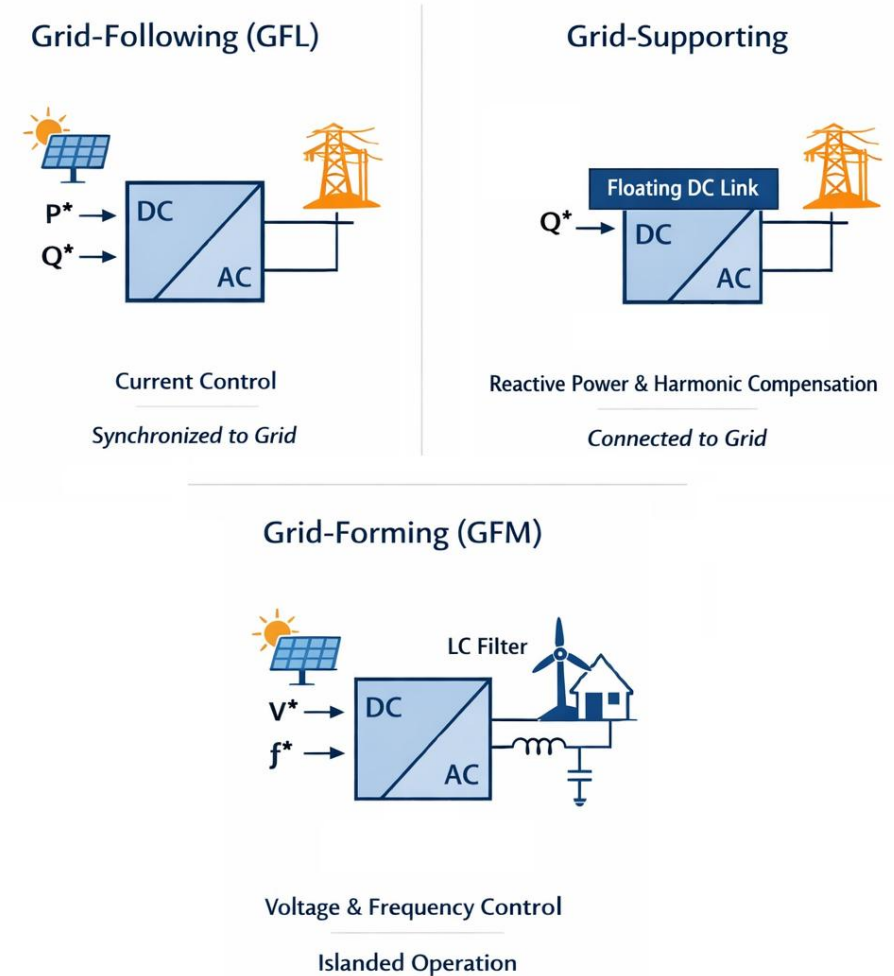


Figure 2.1. Inverter operation modes.

Grid-Following (GFL)

Concept

- Converter behaves as a current-controlled source delivering commanded active and reactive power.

Key Features

- Injects P_{ref} and optionally Q_{ref} .
- Relies on PLL for phase and frequency alignment.
- Inner current loop ensures fast response and good dynamic performance.

Typical Control Structure

- dq -current PI control.
- Outer loop: P/Q regulation.
- Modulation: SPWM / THIPWM / SVPWM.

Applications

- PV inverters, battery inverters operating grid-tied, EV chargers, industrial drives in grid-tie mode.

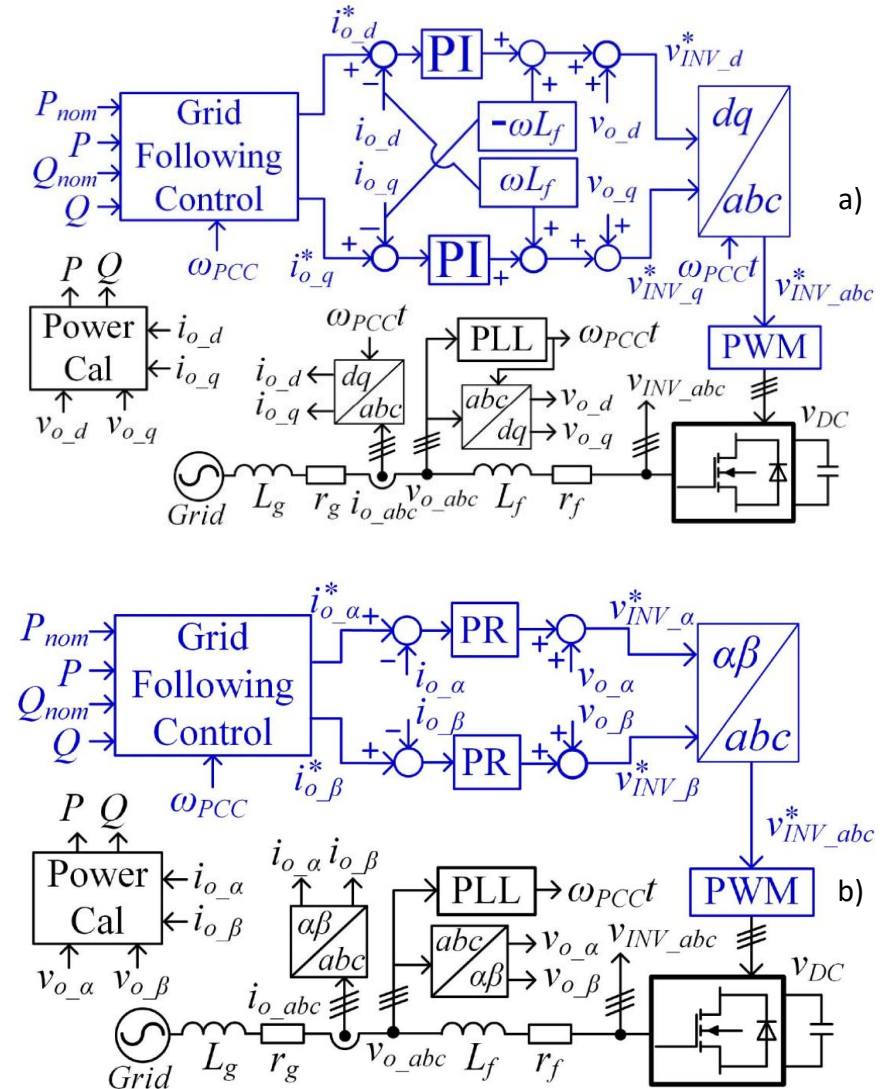


Figure 2.2. Typical Grid Following Control Scheme

a) dq frame, b) $\alpha\beta$ frame. [7]

Grid-Supporting (GSM)

Concept

- Converter operates connected to an existing grid, injecting reactive power and/or harmonic compensation.

Key Features

- Injects Q_{ref} to support grid.
- DC-Link: usually floating capacitor bank to be controlled.
- Supports voltage regulation and power-quality improvement.

Control Structure

- DC-Link voltage control with floating DC-Link.
- Usually current-controlled in dq or $\alpha\beta$.
- May include harmonic resonant controllers (PR or selective harmonic compensators).

Applications

- STATCOMs, active filters, voltage support in weak grids, industrial PQ correction.

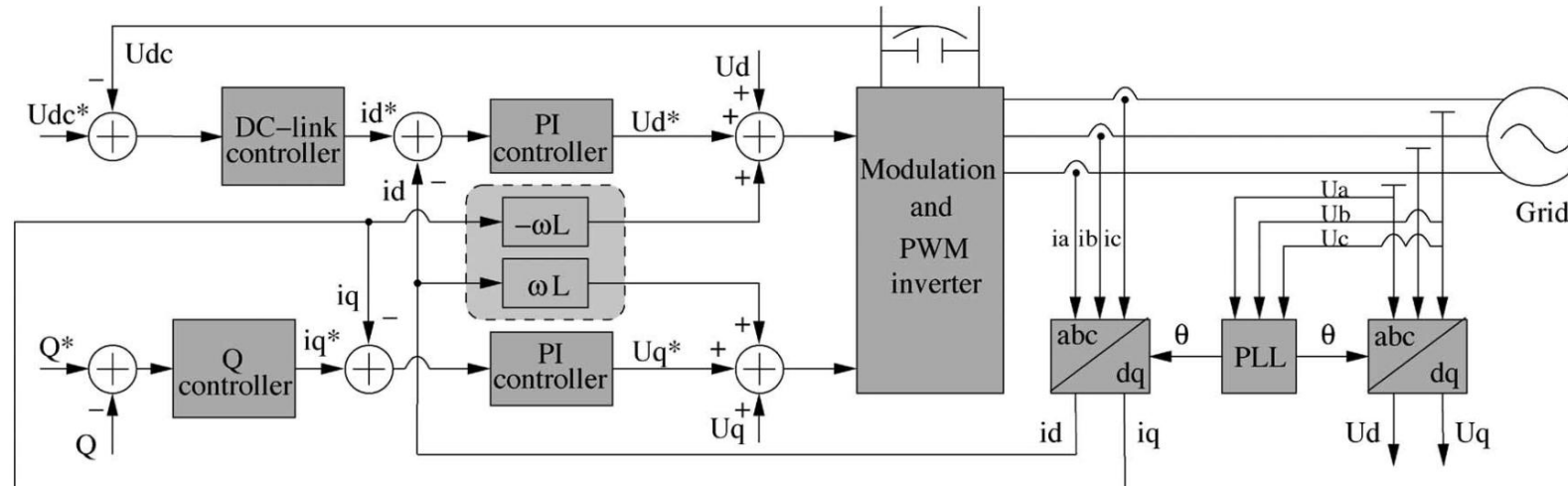


Figure 2.3. General Grid Supporting Control Scheme for PWM Inverter with Floating DC-Link. [1]

Grid Forming, Power Managment

Concept

- GFM unit regulates voltage and frequency using droop laws to share power.
- Mimics synchronous generator behavior: frequency decreases with active power (P–f droop), voltage decreases with reactive power (Q–V droop).

Key Features

- Active power sharing via P–f droop; reactive sharing via Q–V droop. (PD)
- Can be synchronized with a Power Synchronization Loop. (Psync)
- Supports decentralized coordination without communication.
- Smooth response to load changes due to embedded virtual inertia.
- Supports multi-inverter microgrids.
- Works in islanded mode or in weak-grid-connected scenarios.

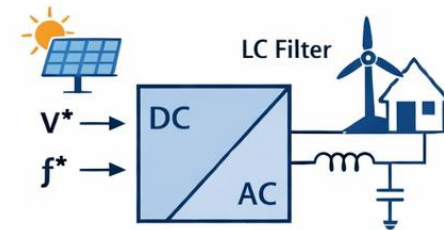
Control Structure

- Droop layer defines f^* and V^* based on P/Q deviations (or Psync)
- Voltage loop + optional current protection loop.
- Compatible with virtual inertia/damping enhancements.

Applications

- Multi-inverter microgrids with shared generation/storage.
- Hybrid RES + BESS requiring coordinated islanded operation.
- Weak grids needing forming capability.

Grid-Forming (GFM)



Voltage & Frequency Control

Grid Forming, Power Droop Mode (GFM-PD): control scheme [6]

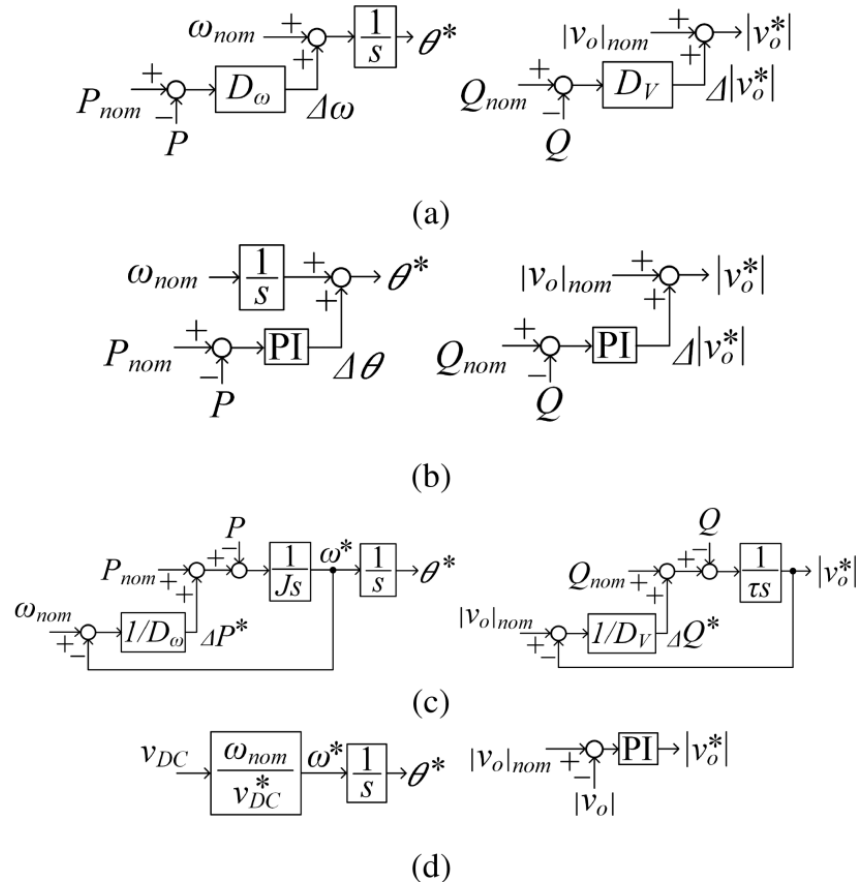


Figure 2.5. Typical grid integration controls of GFM-C. (a) **Droop control.** (b) Power synchronization control. (c) Virtual synchronous generator control. (d) Matching control. [7]

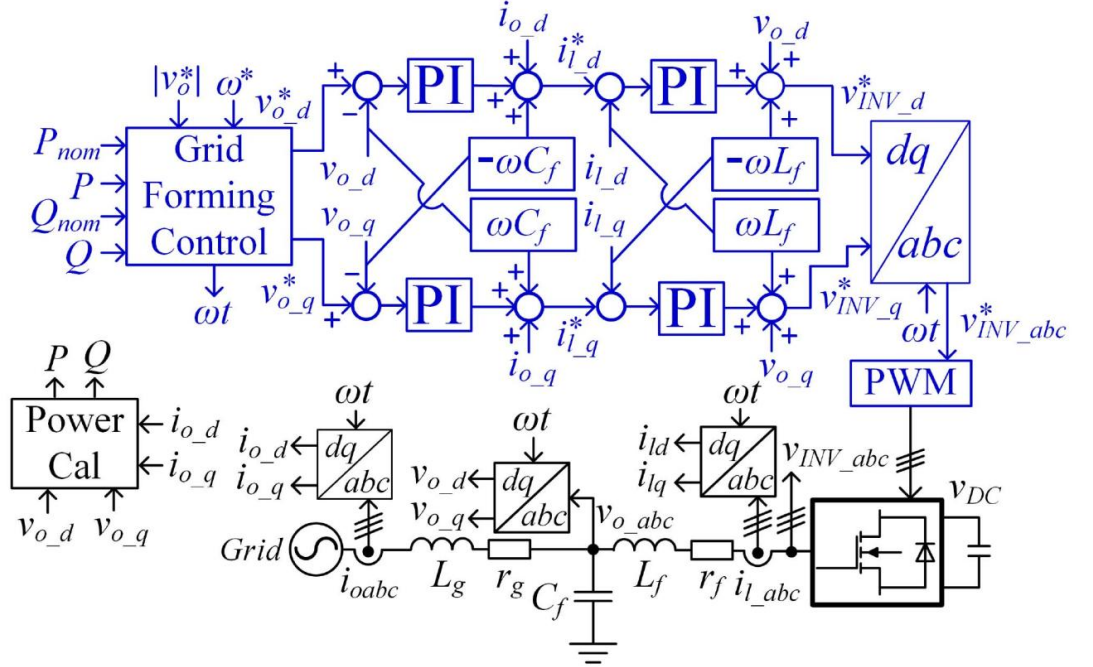


Figure 2.4. Typical converter control loops of Grid Forming Converter. [7]

Grid Forming, Power Synchronization Loop (GFM-PSync): control scheme [5]

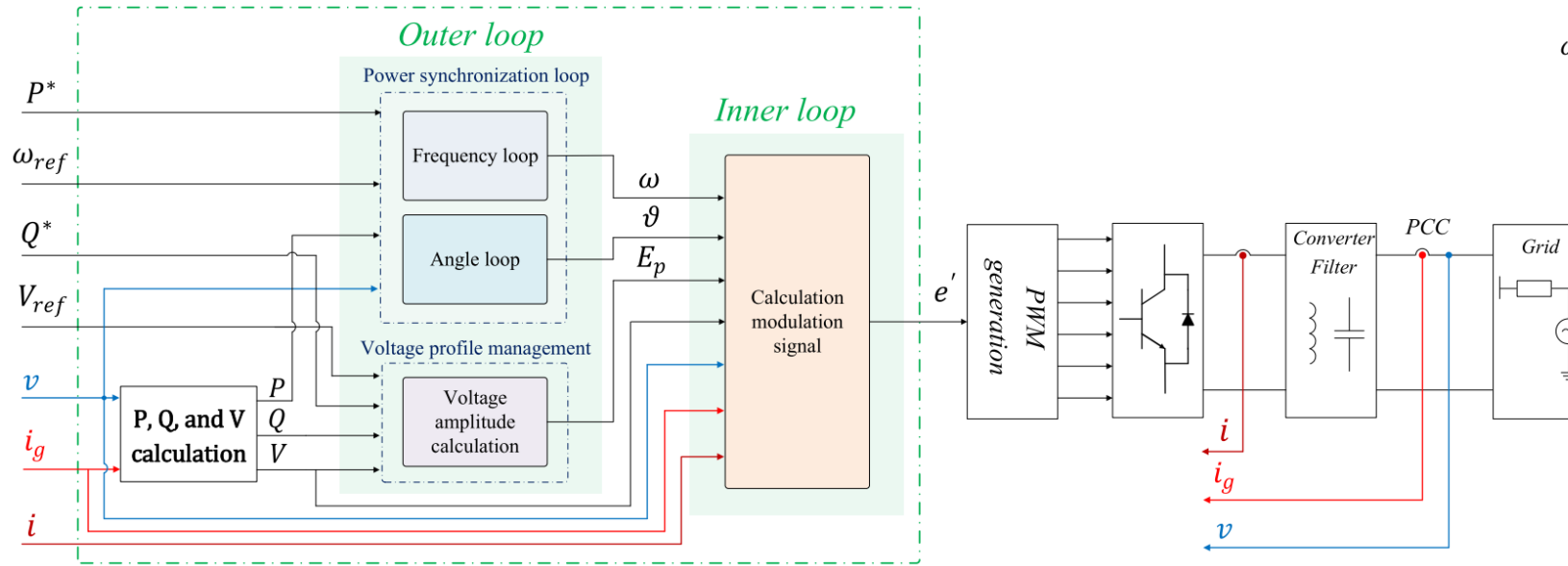


Figure 2.6. Generalized control structure of a GFM converter with Power Synchronization Loop. [6]

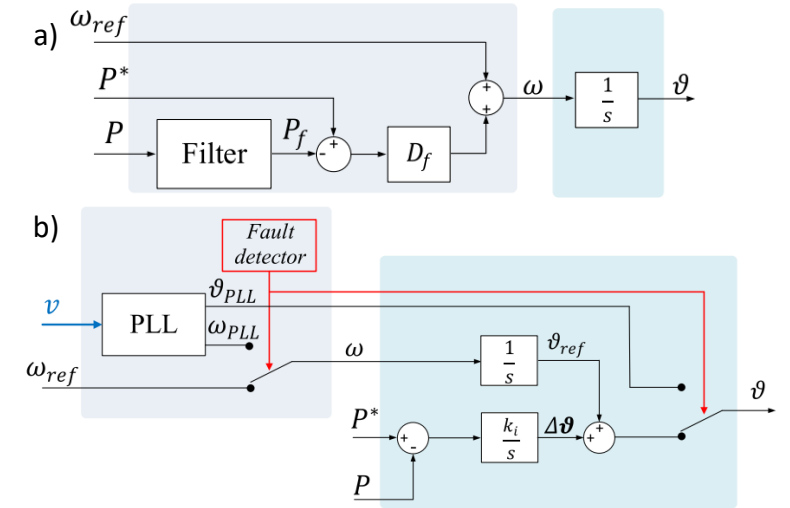


Figure 2.7. Typical power synchronization loops: a) Frequency droop control; b) power control. [6]

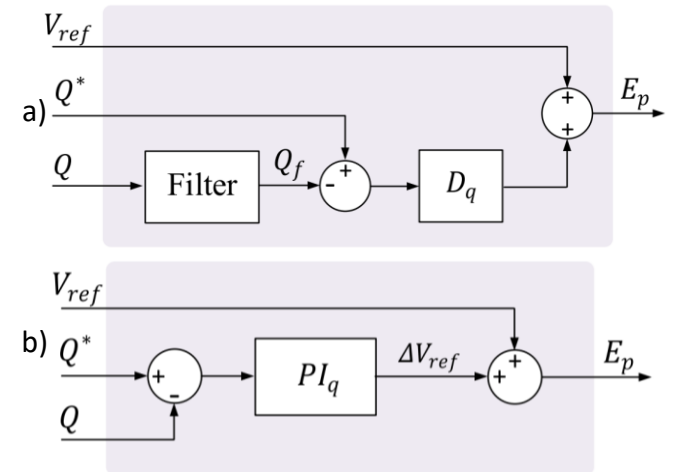


Figure 2.8. Typical voltage profile loops: a) Droop control; b) reactive power control. [6]

Grid Forming, Islanded Mode (GFM-I)

Concept

- Converter acts as an independent voltage source, setting the local voltage and frequency.
- Provides reference for an islanded microgrid without relying on the main grid.

Key Features

- Establishes voltage and frequency autonomously.
- Handles load changes and disturbances with stable voltage behavior.
- Enables black-start and standalone operation

Control Structure

- Outer voltage–frequency loop (forms the grid).
- Output filter capacitor is required to regulate the output voltage.
- Inner current loop for converter protection and shaping.
- Can use droop, VSM, PSC, or VOC-based implementations.

Applications

- Standalone microgrids and UPS systems.
- Black-start units for restoration.
- Battery energy storage systems acting as master unit.

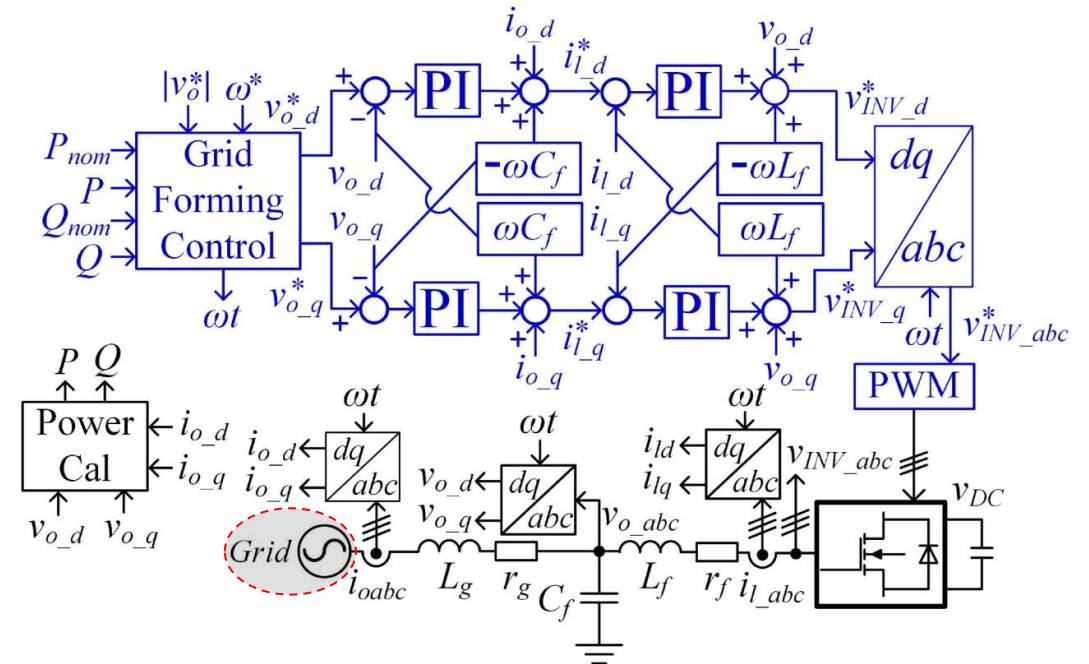


Figure 2.9. Typical converter control loops of Grid Forming Converter. [6]
In islanded mode there is no connection to an existing electrical grid, and the converter creates the grid.

Presentation references



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